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Interpreting hominid behavior on the basis of sexual dimorphism

Numerous studies use estimates of sexual dimorphism in canine tooth size and body weight to support speculation about the behavior of australopithecines. However, the range of mating systems inferred for australopithecines encompasses virtually the entire spectrum of mating systems seen among extant anthropoid primates, from monogamy to polygyny characterized by intense male–male competition. This variety of opinion can be attributed partly to the unusual combination of high body size dimorphism and reduced canine dimorphism in australopithecines. Here we provide a joint comparison of recent models for the behavioral correlates of both canine dimorphism and body size dimorphism, and apply this to published estimates of dimorphism in body size and canine tooth size in hominids. Among extant species, body weight dimorphism and canine dimorphism are strongly correlated with estimates of intrasexual competition. Canine crown height dimorphism provides the best discrimination between taxa that show high degrees of male–male competition, and those that do not. Relative male maxillary canine tooth size offers additional evidence about male–male competition. On the other hand, canine occlusal dimorphism offers little discrimination among species of different male–male competition levels. Estimates of canine dimorphism, relative canine size, and body weight dimorphism in australopithecines provide little definitive information about male–male competition or mating systems. Dimorphism of *Australopithecus africanus* and *Australopithecus robustus* can be reconciled with a mating system characterized by low-intensity male–male competition. The pattern of dimorphism and relative canine size in *Australopithecus afarensis* and *A. robustus* provides contradictory evidence about mating systems and male–male competition. We review a number of hypotheses that may explain the unusual pattern of dimorphism of *A. afarensis* and *Australopithecus boisei*, but non-satisfactorily resolves the problem given current data.

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Introduction

Anthropologists have long speculated about the evolution of human mating systems. Because sexual dimorphism in canine tooth size and body size is strongly associated with variation in intrasexual competition in extant primates, and hence mating system (Kay *et al.*, 1988; Greenfield, 1992*a,b*; Plavcan, 1990; Plavcan & van Schaik, 1992, 1994; Plavcan *et al.*, 1995), it has long been used as evidence for making inferences about hominid mating systems.

There is a general consensus that all australopithecines were characterized by relatively slight canine tooth size dimorphism (Lovejoy, 1981; Leutenegger & Shell, 1987; McHenry, 1992, 1994*a,b*), a character state that is often recognized as a derived feature of the hominid lineage (e.g. White *et al.*, 1994). Estimates of body size dimorphism are more variable, particularly for *Australopithecus afarensis* (Lovejoy *et al.*, 1989; Jungers, 1988; McHenry, 1986, 1988, 1991, 1992, 1994*a,b*; Kimbel & White, 1988; Hartwig-Scherer, 1993; Cole & Smith, 1987; Johanson *et al.*, 1978; Johanson *et al.*, 1982; Stern & Susman, 1983; Susman *et al.*, 1984). Where body size dimorphism is thought to be comparatively high, the dissociation in the magnitude of sexual dimorphism in these two traits presents a paradox leading to a dichotomy of inferences about mating systems. On the one hand, strong body size dimorphism suggests a polygynous mating system characterized by high levels of male–male competition (Wolpoff, 1976; Leutenegger & Shell, 1987; Foley & Lee, 1989; McHenry, 1994*a,b*). On the other hand,

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relatively weak canine tooth size dimorphism suggests either monogamy (Lovejoy, 1981), or a mating system characterized by low levels of male–male competition as seen in, for example, *Brachyteles* or *Pan* (Kelley, 1989).

Several studies inferring polygyny on the basis of strong body size dimorphism attempt to resolve this conflict by providing post hoc explanations for why canine dimorphism should be so low. The most often cited of these is the ''weapons-replacement'' hypothesis of Darwin (1871; Wolpoff, 1976; McHenry, 1992, 1994*a*,*b*), which posits that weapons replaced the canines in intrasexual competition, thereby eliminating the selective value of canines as weapons. Wolpoff (1976) recognizes a weakness in this argument, in that it does not explain why weapons reduce the selective value of large canine teeth, but not of large body size. He suggests that large male body size is favored because larger males should be able to more effectively wield hand-held weapons. On the other hand, McHenry (1994*a*,*b*) suggests that male *A. afarensis* had relatively large arms because males used their arms, rather than their canine teeth, in fights. Others suggest mechanisms for canine reduction that do not involve the substitution of other structures for use as weapons. For example, Jolly (1970) suggests that canines were reduced because they inhibited jaw movements during chewing, Leutenegger & Shell (1987) suggest that canine reduction occurred as a result of dental crowding associated with a reduction in facial prognathism, while most recently Greenfield (1990, 1992*a*) suggests that canines were reduced in association with selection for their incorporation into the incisal functional field (following a reduction in selection for their use as weapons).

Only Leutenegger & Kelly (1977) and Leutenegger (1982) have directly compared body size dimorphism and canine tooth size dimorphism in primates. They found that dimorphism in both traits is correlated, but suggested that each is independently affected by different factors, resulting in dissociation of the two traits in several species. Beyond this, no study has compared the predictive power of both types of dimorphism for inferring behavior in the fossil record. Given the recent proliferation of studies on the causes of dimorphism in canine tooth size (Harvey *et al.*, 1978; Lucas *et al.*, 1986; Leutenegger, 1982; Kay *et al.*, 1988; Plavcan, 1990, 1993; Greenfield, 1990, 1992*a*,*b*; Plavcan & van Schaik, 1992, 1994; Plavcan *et al.*, 1995) and body size (Clutton-Brock *et al.*, 1977; Leutenegger & Cheverud, 1982, 1985; Cheverud *et al.*, 1985; Gaulin & Sailer, 1984; Ely & Kurland, 1989; Ford, 1994; Plavcan & van Schaik, 1997), such an analysis is overdue. In particular, a framework needs to be established for evaluating the type and strength of inferences about hominid social behavior that can be drawn by comparing patterns of both canine tooth size dimorphism and body weight dimorphism between extinct and extant species.

Here, we compare variation in canine tooth size dimorphism, relative male canine tooth size, and body weight dimorphism to estimates of male–male competition, and evaluate the behavioral implications of inferred patterns of dimorphism in *A. afarensis*, *Australopithecus africanus*, *Australopithecus boisei*, and *Australopithecus robustus*. Our intention is not to provide definitive statement about the behavior of australopithecines. Rather, we seek to evaluate what can—and cannot—be said about the social behavior of these hominids on the basis of comparisons to dimorphism in extant primates.

Materials and methods

Dental data

Dental metric data (Table 1) on male and female canine tooth size for 84 extant anthropoid species are taken from Plavcan (1990). We evaluate six canine tooth size measures: mesiodistal

Table 1 Male and female body weights (in kg), dimorphism estimates and competition levels used in this analysis

Species	Weight M	Weight F	Weight dim	Mandibular dim MD	Mandibular dim BL	Mandibular dim Height	Maxillary dim MD	Maxillary dim BL	Maxillary dim Height	Male MD	Male BL	Male Height	Comp
Cercocebus albigena	8·98	6·40	1·40	1·25	1·25	1·66	1·39	1·20	1·89	8·25	5·69	16·73	4
Cercocebus galeritus	10·18	5·47	1·86	—	—	—	—	—	—	—	—	—	4
Cercocebus torquatus	11·41	6·33	1·80	1·44	1·38	2·00	1·60	1·41	2·46	10·35	7·04	23·53	4
Cercopithecus aethiops	4·58	3·26	1·40	1·28	1·20	1·63	1·42	1·29	1·80	7·37	5·14	17·66	4
Cercopithecus ascanius	3·71	2·96	1·25	1·23	1·18	1·45	1·29	1·15	1·56	6·80	4·51	16·13	3
Cercopithecus cephus	4·09	2·88	1·42	1·30	1·23	1·53	1·42	1·22	1·77	6·89	4·62	16·32	3
Cercopithecus lhoesti	5·97	3·45	1·73	1·43	1·41	1·75	1·49	1·38	1·84	8·39	5·76	19·93	3
Cercopithecus mitis	5·85	3·93	1·49	1·43	1·38	1·81	1·58	1·42	1·97	8·29	5·62	19·24	3
Cercopithecus mona	4·40	2·50	1·76	1·39	1·41	1·85	1·46	1·32	1·90	7·82	5·03	17·76	3
Cercopithecus neglectus	6·32	3·96	1·60	1·37	1·36	1·51	1·50	1·41	1·71	8·56	5·72	20·02	3
Cercopithecus nicititans	6·73	4·28	1·57	1·36	1·33	1·56	1·53	1·33	1·61	8·50	5·60	18·10	3
Cercopithecus pogonias	4·50	3·02	1·49	1·46	1·26	1·29	1·33	1·23	1·55	7·01	4·65	15·73	3
Colobus angolensis	9·62	7·77	1·24	1·27	1·26	1·78	1·31	1·32	1·84	9·30	6·66	17·66	4
Colobus badius	9·35	6·77	1·38	1·29	1·23	1·74	1·22	1·23	1·88	8·30	6·28	17·60	4
Colobus guereza	9·33	7·83	1·19	1·15	1·11	1·38	1·20	1·08	1·46	9·84	6·71	20·31	3
Colobus polykomos	9·90	8·30	1·19	1·07	1·13	1·65	1·14	1·08	1·76	9·20	6·68	19·13	4
Colobus satanas	12·00	9·50	1·26	1·12	1·08	1·72	1·15	0·99	1·75	7·86	6·20	16·08	4
Colobus verus	4·60	4·10	1·12	1·36	1·22	1·81	1·37	1·20	2·00	7·19	5·58	14·90	4
Erythrocebus patas	11·10	5·90	1·88	1·44	1·43	1·73	1·53	1·49	2·17	10·58	6·72	26·50	3
Gorilla gorilla	159·20	97·70	1·63	1·41	1·38	1·64	1·56	1·48	1·73	23·76	16·80	30·26	3
Homo sapiens	47·90	40·10	1·20	1·04	1·07	1·10	1·12	1·07	1·08	9·29	8·40	10·85	–
Hylobates concolor	5·60	5·80	0·96	1·03	1·07	1·21	1·05	1·03	1·15	8·14	5·31	19·11	1
Hylobates hoolock	6·90	6·50	1·06	1·04	1·09	1·14	1·09	1·05	1·03	8·42	5·92	17·75	1
Hylobates klossi	5·70	5·90	0·97	1·02	1·01	0·98	1·00	0·99	0·94	7·71	5·41	15·72	1
Hylobates lar	5·71	5·30	1·08	1·07	1·12	1·15	1·13	1·12	1·16	7·92	5·35	18·32	1
Hylobates syndactylus	10·85	10·60	1·02	1·08	1·13	1·14	1·15	1·10	1·18	9·72	6·68	20·94	1
Macaca arctoides	12·20	8·40	1·45	—	—	—	—	—	—	—	—	—	4
Macaca fascicularis	4·54	3·12	1·46	1·39	1·46	1·60	1·56	1·41	2·25	9·73	6·43	24·09	4
Macaca fuscata	11·70	9·10	1·28	1·45	1·35	1·62	1·54	1·23	2·03	10·20	6·46	19·57	4
Macaca mulatta	6·80	5·18	1·31	1·46	1·41	1·67	1·41	1·38	2·07	8·67	6·42	16·97	4
Macaca nemestrina	10·21	6·35	1·61	1·51	1·42	1·95	1·65	1·35	2·36	12·29	7·80	28·89	4
Macaca nigra	7·99	4·69	1·70	1·53	1·42	1·73	1·70	1·51	2·61	11·55	7·95	29·73	4
Macaca silenus	6·80	5·00	1·36	1·51	1·43	2·06	1·76	1·53	2·43	11·04	7·45	24·66	4
Macaca sinica	6·50	3·40	1·91	1·29	1·39	1·49	1·56	1·30	1·97	9·05	6·38	19·69	4
Mandrillus leucophaeus	17·00	10·00	1·70	1·73	1·76	2·85	2·08	1·76	4·10	17·89	11·02	47·97	4
Miopithecus talapoin	1·38	1·12	1·23	1·22	1·28	1·54	1·30	1·18	1·69	4·83	3·56	11·35	4
Nasalis larvatus	21·20	10·00	2·12	1·20	1·46	1·89	1·33	1·30	2·20	9·23	6·75	22·96	4
Pan paniscus	45·00	33·20	1·36	1·17	1·18	1·31	1·22	1·28	1·38	11·44	8·88	15·6	2
Pan troglodytes	43·00	33·20	1·30	1·23	1·32	1·26	1·44	1·31	1·43	15·30	11·80	21·72	2
Papio anubis	25·10	14·10	1·78	1·55	1·55	1·81	1·82	1·66	2·22	16·10	11·57	35·43	4
Papio cynocephalus	17·24	9·75	1·77	1·43	1·45	2·44	1·50	1·47	2·80	9·72	7·16	25·57	4
Papio hamadryas	19·58	9·98	1·96	1·61	1·75	2·16	1·80	1·68	2·74	14·14	10·66	30·64	4
Papio ursinus	27·45	14·52	1·89	1·62	1·68	2·83	1·95	1·75	3·83	17·00	10·72	46·53	4
Pongo pygmaeus	86·30	38·70	2·23	1·32	1·47	1·55	1·31	1·36	1·69	17·50	14·26	27·00	3
Presbytis comata	6·68	6·66	1·00	1·09	1·06	1·50	1·13	1·00	1·50	6·75	5·00	13·93	3
Presbytis cristatus	6·60	5·70	1·16	1·10	1·12	1·53	1·19	1·10	1·69	8·12	6·39	18·57	3
Presbytis entellus	10·64	6·70	1·59	1·31	1·29	1·76	1·42	1·26	1·88	9·58	6·87	19·80	4
Presbytis johnii	12·70	10·90	1·16	1·37	1·27	1·73	1·41	1·21	1·63	3·81	4·62	9·31	3
Presbytis melalophos	6·48	6·32	1·02	1·03	1·04	1·46	1·10	0·91	1·71	6·40	4·51	13·95	3
Presbytis obscurus	7·39	6·47	1·14	1·13	1·27	1·55	1·24	1·07	1·77	7·62	5·75	15·10	3
Presbytis pileatus	5·77	5·14	1·12	1·12	1·17	1·43	1·31	1·13	1·88	9·09	6·66	19·85	3
Presbytis potenziani	6·50	6·40	1·02	0·99	1·10	1·42	1·08	0·94	1·41	7·25	5·33	13·56	1

Table 1 continued on next page

Table 1 ***Continued from previous page***

Species	Weight M	Weight F	Weight dim	Mandibular dim MD	Mandibular dim BL	Mandibular dim Height	Maxillary dim MD	Maxillary dim BL	Maxillary dim Height	Male MD	Male BL	Male Height	Comp
Presbytis rubicunda	6·20	5·70	1·09	1·01	1·08	1·37	1·03	0·94	1·50	5·89	4·20	12·73	3
Presbytis vetulus	5·67	5·11	1·11	1·20	1·79	1·61	1·27	1·19	1·81	7·82	6·04	16·96	3
Theropithecus gelada	19·00	11·70	1·62	1·62	1·62	2·16	1·89	1·61	3·22	14·89	9·32	39·62	4
Alouatta belzebul	7·18	5·18	1·39	1·35	1·46	1·78	1·52	1·37	1·81	8·03	6·06	12·28	4
Alouatta caraya	8·28	5·41	1·53	1·31	1·36	1·49	1·38	1·30	1·49	8·58	5·60	14·24	4
Alouatta fusca	6·22	4·53	1·37	1·37	1·48	1·63	1·41	1·37	1·51	8·13	5·83	13·18	4
Alouatta palliata	7·16	5·60	1·28	1·28	1·32	1·46	1·26	1·26	1·51	8·16	5·81	13·79	4
Alouatta pigra	11·59	6·29	1·84	1·25	1·24	1·49	1·26	1·28	1·37	8·48	5·92	13·94	4
Alouatta seniculus	7·62	6·02	1·26	1·20	1·35	1·35	1·30	1·39	1·50	8·64	6·65	13·42	4
Aotus lemurinus	0·89	0·92	0·97	1·02	1·07	1·10	1·02	0·99	1·17	3·19	2·69	5·62	1
Aotus trivirgatus	0·92	0·93	0·99	1·04	1·05	1·06	1·07	1·01	1·08	3·34	2·61	5·19	1
Ateles geoffroyi	8·38	6·70	1·25	1·14	1·19	1·40	1·12	1·01	1·52	6·38	5·17	11·43	2
Ateles paniscus	9·45	8·59	1·10	1·07	1·16	1·37	1·11	1·07	1·57	6·60	5·29	12·12	2
Brachyteles arachnoides	9·61	8·38	1·15	1·21	1·24	1·46	1·10	1·08	1·33	6·85	5·82	9·47	2
Cacajao calvus	3·45	2·88	1·20	1·15	1·22	1·20	1·15	1·24	1·26	8·15	7·26	13·58	3
Callicebus brunneus	0·84	0·85	0·99	—	—	—	—	—	—	—	—	—	1
Callicebus moloch	1·08	0·99	1·09	1·03	0·98	1·08	0·98	1·00	1·08	2·84	2·75	4·36	1
Callicebus personatus	1·27	1·38	0·92	—	—	—	—	—	—	—	—	—	1
Callicebus torquatus	1·10	1·31	0·84	0·99	1·02	1·02	1·01	1·03	1·04	3·50	3·18	4·28	1
Callithrix jacchus	0·21	0·24	0·87	0·99	1·04	1·07	1·05	1·04	1·03	2·52	2·08	5·11	1
Cebuella pygmaea	0·11	0·12	0·92	1·02	0·99	0·99	1·01	0·99	1·03	1·64	1·34	3·09	1
Cebus apella	3·37	2·45	1·38	1·23	1·29	1·42	1·19	1·18	1·41	7·49	6·45	13·67	4
Cebus olivaceous	2·23	1·59	1·40	—	—	—	—	—	—	—	—	—	3
Chiropotes satanas	2·88	2·66	1·08	1·05	1·07	1·16	1·06	1·10	1·20	6·59	5·70	11·18	2
Lagothrix lagothricha	6·67	5·54	1·20	1·15	1·24	1·56	1·29	1·21	1·77	7·43	6·54	14·23	2
Leontopithecus rosalia	0·51	0·48	1·06	1·02	1·12	1·31	1·01	1·06	1·18	3·44	3·02	6·33	1
Pithecia pithecia	1·80	1·51	1·19	1·17	1·16	1·26	1·17	1·20	1·32	5·00	4·60	8·93	2
Saimiri boliviensis	1·10	0·70	1·57	1·13	1·28	1·67	1·23	1·04	1·61	3·54	2·92	6·66	3
Saimiri oerstedi	0·75	0·60	1·25	1·13	1·41	1·67	1·33	1·11	1·68	3·81	3·30	7·58	3
Saimiri sciureus	0·96	0·79	1·22	1·10	1·22	1·55	1·21	1·15	1·41	3·94	3·26	7·10	3
Saguinus fuscicollis	0·41	0·41	1·00	0·97	0·98	0·98	0·98	1·00	1·03	2·77	2·31	5·66	1
Saguinus midas	0·49	0·52	0·94	1·00	0·98	0·98	0·99	1·03	0·99	2·80	2·31	5·38	1
Saguinus mystax	0·57	0·58	0·98	—	—	—	—	—	—	—	—	—	1
Saguinus labiatus	0·48	0·52	0·92	—	—	—	—	—	—	—	—	—	1
Saguinus oedipus	0·48	0·50	0·96	1·01	1·03	0·98	1·00	1·04	1·00	2·90	2·42	5·65	1

MD, mesiodistal; BL, buccolingual; Dim, dimorphism estimates; Comp, competition levels.

and buccolingual lengths measured in the occlusal plane at the base of the maxillary and mandibular canines, and the crown height of each canine measured from the apex of the tooth to the cementum–enamel junction.

We used several sources for hominid dental data. Canine dimensions of *A. afarensis* were taken from White (1977) and Johanson *et al.* (1982). We used data kindly supplied by Fred Grine for *A. africanus* and *A. robustus* (most of these are published in Grine 1989, 1993). Several unpublished specimens from Sterkfontein were also included with permission from P. V. Tobias. Because these additional data are not generally available, we repeated calculations for the published and unpublished data sets. Results did not substantially differ. White *et al.* (1981) provide means, ranges, sample sizes, and standard deviations of occlusal canine dimensions for large samples of *A. afarensis*, *A. africanus*, *A. robustus*, and *A. boisei*. We used the CV method (see below) to check our estimates of dimorphism for *A. africanus* and *A. robustus* with those based on

this larger data set. In most cases, results did not substantially differ (we note in the text where they did). Finally, we used summary data of occlusal dimensions of *A. boisei* published in White *et al.* (1981).

Most literature reports do not provide data on the crown height of the canine teeth of australopithecines. However, recent comparative studies of canine dimorphism suggest that crown height is the most important aspect of the canine tooth in its function as a weapon, and that canine crown height dimorphism is strongly correlated with behavioral and ecological variables (Greenfield, 1990, 1992; Plavcan, 1990, 1993; Plavcan & van Schaik, 1992, 1994; Plavcan *et al.*, 1995). For this reason data on maxillary canine crown heights is especially valuable for evaluating hominid behavior on the basis of canine dimorphism.

Data on maxillary canine crown height for five relatively unworn specimens of *A. africanus* (Stw 252, Sts 52, and MLD 11, Stw 287, Stw 369, Stw 498), and five specimens of *A. robustus* (Sk 4, Sk 27, Sk 55, Skx 162, and Skx 25296) were kindly provided by Fred Grine. These measurements were checked by J.M.P. against casts housed at the State University of New York at Stony Brook. Grine's measurements were preferred because they were taken from the original specimens. Casts of 16 additional maxillary canines were also examined, though all showed too much apical wear for meaningful measurements of crown height. Maxillary canine crown height measurements of four relatively unworn specimens of *A. afarensis* (LH 3, LH 6, AL 200-1a, and AL 333x-3) were kindly provided by Jay Kelley. Casts of LH 5, AL 199-a, AL 333-2, AL 333-1 were also examined, but maxillary canines were too worn to provide accurate measurements of crown height. No unworn canines of *A. boisei* were available to us that could yield meaningful crown height measurements.

Body weight data

Male and female body weights for 86 extant anthropoid species (Table 1) were taken from Plavcan & van Schaik (1997). These body weight data were taken for the most part from primary literature reports and museum specimens. All data were for wild specimens only. Wherever possible, data from a single subspecies was used. Details about the data, and the justifications for using individual values are provided in Plavcan & van Schaik (1997). Because both body weights and canine measurements were not available for either males or females for a number of species, sample sizes vary between comparisons of canine dimorphism and body weight dimorphism.

Estimating hominid body weights and body weight dimorphism, particularly for *A. afarensis*, has been and continues to be debated in the recent literature (McHenry, 1992; Hartwig-Scherer, 1993; Richmond & Jungers, 1995; Lockwood *et al.*, 1996). For our analyses, we rely on McHenry (1992), who extrapolates body weights from lower limb and coaxial elements using regression formulae derived from a series of extant hominoids and humans. For each hominid, McHenry provides two body weight estimates (one based on hominoid regressions, the other on human regressions). We employ both estimates in our analyses (Table 2). McHenry's (1992) estimates of weight dimorphism are lower than those presented by McHenry (1988). McHenry (1992) points out that the taxonomic assignment of the elements used in his analysis is not certain in some cases, especially *A. boisei* (McHenry, 1992).

It is important to remember that McHenry (1992) presents only body weight *estimates*, not actual body weights. Virtually all discussion of the implications of size dimorphism in hominids centers on comparisons with body weight dimorphism in extant taxa, possibly because most comparative analyses of the behavioral and ecological correlates of dimorphism in extant species focus on body weight. Obviously, hominid dimorphism ideally should be compared

Table 2 Estimates of sexual dimorphism and ''maximum'' canine tooth size in hominids*

	Weight			Mandibular dim			Maxillary dim			Male canine (mm)		
Species	M	F	Dim	MD	BL	Height	MD	BL	Height	MD	BL	Height
Australopithecus afarensis												
Mean method	(6) 44·60	(4) 29·40	1·52	(6) 1·25	(10) 1·22	—	(10) 1·11	(10) 1·19	(4) 1·19	11·60	12·50	15·60
CV method			1·53	1·34	1·25	—	1·12	1·18	1·18			
Mean method	60·10	35·60	1·69									
CV method			1·70									
Australopithecus africanus†												
Mean method	(5) 40·80	(7) 30·80	1·32	(30) 1·09	(30) 1·13	(10) 1·15	(11) 1·15	(11) 1·23	(5) 1·13	10·00	10·30	16·10
CV method			1·39	1·08	1·14	1·14	1·14	1·22	1·13			
Mean method	52·80	36·20	1·46									
CV method			1·57									
Australopithecus robustus‡												
Mean method	(2) 40·20	(2) 31·90	1·26	(13) 1·16	(12) 1·18	—	(14) 1·13	(15) 1·17	(5) 1·25	9·20	11·20	16·00
CV method			1·32	1·15	1·21	—	1·13	1·15	1·31			
Mean method	49·80	40·30	1·24									
CV method			1·53									
Australopithecus boisei												
Mean method	(1) 48·60	(1) 34·00	1·43	—	—	—	—	—	—	10·80	9·90	—
CV method			—	(5) 1·08	—	—	(8) 1·28	(4) 1·25				
Mean method	76·00	42·00	1·81									
CV method			—									

*For the hominids, body weights are values reported by McHenry (1992) from separate regressions using humans (first set of weight estimates) and hominoids (second set of weight estimates). Values of canine dimorphism were calculated using data from Johanson *et al.* (1982) and White (1977) for *A. afarensis*, White *et al.* (1981) for *A. boisei*, and data supplied by Fred Grine for *A. africanus* and *A. robustus*. Dimorphism estimates from Grine's data are comparable to values calculated from Johanson *et al.* (1982). Values of dimorphism for the ''mean method'' and ''CV method'' are calculated using the methods of Plavcan (1994). Sample sizes for data used to calculate dimorphism estimates are in parentheses. The values for maxillary canine size are simply the largest published dimension available from all sources.

†For *A. africanus*, data from White (1977) and Johanson *et al.* (1982) yield values of 1·06 and 1·07 for the maxillary MD and BL dimensions, respectively, using the CV method.

‡For *A. robustus*, data from White (1977) and Johanson *et al.* (1982) yield a value of 1·06 for the maxillary MD dimensions using the CV method.

Dim, dimorphism estimates; MD, mesiodistal; BL, buccolingual; CV, coefficient of variation.

with variation in *skeletal* dimorphism in extant taxa. However, analyses comparing skeletal dimorphism in extant primates with behavioral and ecological variables correlated with variation in weight dimorphism have not yet been done, and are beyond the scope of this analysis. The purpose of our analysis is not to fill this gap, but rather to evaluate what can and cannot be said about estimates of weight dimorphism in hominids.

Sexual dimorphism

We estimated sexual dimorphism of extant species as the natural logarithm of the male mean divided by the female mean (Table 1). This is equivalent to subtracting the natural logarithm of the female mean from that of the male mean, and reduces skewness and kurtosis of the data across species (Gaulin & Sailer, 1984; Plavcan, 1990; Plavcan & van Schaik, 1992).

Because most fossil specimens are fragmentary, the sex of most hominid specimens cannot be reliably determined on the basis of discrete, sex-specific traits. Consequently, prior studies quantify hominid dimorphism in one of two ways. McHenry (1992) simply calls large specimens males, and small specimens females. This is, in most cases, equivalent to the ''mean'' method of Godfrey *et al.* (1993) and Plavcan (1994), in which a sample is divided into large and small specimens about the combined sample mean, and dimorphism is estimated as the ratio of large (''male'') divided by small (''female'') specimens. This method tends to overestimate sexual dimorphism when the true sample dimorphism is low (the exact level depends on the degree of intrasexual variability in the sample), but is otherwise quite accurate when dimorphism is more substantial. An alternative method is extrapolation of sexual dimorphism from coefficients of variation (CVs—Pilbeam & Zwell, 1972; Kay, 1982; Leutenegger & Shell, 1987; Plavcan, 1994). Like the mean method, this method tends to overestimate sexual dimorphism when dimorphism is low (less than a ratio value of about 1.3). Additionally, this method is less reliable when intrasexual variability in a trait is high (traits showing CVs greater than about 10). The CV method is useful for estimating dimorphism in teeth because dental intrasexual variability tends to be low (Gingerich & Schoeninger, 1979; Plavcan, 1990, 1993; Cope, 1993). However, intrasexual variability of skeletal elements tends to be comparatively high, meaning that the CV estimates of weight dimorphism from skeletal elements are probably overestimated.

For comparison, we calculated estimates of dimorphism for all measures (body weight and canine tooth size) using both the CV and mean methods (Table 2). For body weight dimorphism, estimates derived from the CV method are higher than those derived from the mean method. For the reasons noted above, we rely on the mean method estimates of weight dimorphism in our analysis. For the canine dimensions, estimates of dimorphism from both methods are very similar in most cases. For most estimates, we simply chose the larger of the two for evaluation (rendering the analysis conservative). Where estimates differed substantially, we evaluated both. Additionally, Grine's data yields substantially higher estimates of dimorphism than data from White (1977) and White *et al.* (1981) for maxillary mesiodistal (MD) and buccolingual (BL) dimensions of *A. africanus*, and the maxillary MD dimension of *A. robustus* (see the footnote in Table 2). In these cases, we evaluated all estimates of dimorphism.

Relative canine tooth size

A number of recent studies emphasize that sexual dimorphism is a function of relative changes in both male and female traits (Shea, 1986; Leigh & Shea, 1995; Plavcan & van Schaik, 1994; Plavcan *et al.*, 1995). Plavcan (1993) and Plavcan *et al.* (1995) have demonstrated a link between both male and female relative maxillary canine tooth size, and categorical estimates of

intrasexual agonistic competition. Thus, more accurate estimates of behavior potentially could be gleaned for extinct taxa by considering dimorphism as a function of the independent development of male and female traits. Unfortunately, two factors make such comparisons difficult for fossil taxa. First, for both the canines and body weight estimates, the sex of individual specimens cannot be reliably determined. Second, for body weight, it is difficult to simultaneously control for interspecific differences in body weight and determine relative changes in male and female body weight that produce sexual dimorphism.

While these problems preclude a comparison of separate male and female components of body weight dimorphism in hominids, it is still possible to carry out a limited comparison of relative canine tooth size in fossil hominids with that among extant species. As a conservative approach, we simply compare the largest relative canine size for all hominid specimens with that of extant male primates. Given enough specimens (greater than 6), there is a high probability that both sexes are present in the sample, and that the largest canine provides a conservative upper limit for the average canine size of the species as a whole. Where the sample size is less than 6 (as for the canine crown height dimensions), interpretations about relative canine size must be made under the explicit assumption that at least one male is present in the sample.

Relative canine tooth size was calculated as the residual from a least-squares regression of maxillary canine tooth size versus body weight. We used estimated average male body weights for the hominids, repeating the analyses for maximum and minimum estimated weights reported by McHenry (1992). For canine crown height, we calculated separate residuals for each specimen (four *A. afarensis*, five *A. africanus*, five *A. robustus*). For the occlusal canine dimensions of each australopithecine species, we used the largest value of each dimension available from the sources noted above. Finally, all of the analyses were repeated calculating residuals from an isometric line passed through the data.

Male–male competition

Sexual dimorphism has long been used as evidence of mating system or intrasexual competition in hominids and other extinct species. Here, we examine the relation between sexual dimorphism and the male–male ''competition levels'' (Table 1) of Plavcan & van Schaik (1992).

Each extant species is classified into one of four competition levels (labeled 1 through 4) primarily on the basis of categorical estimates of the ''intensity'' and ''potential frequency'' of male–male competition. The intensity of male–male competition is dichotomized into a ''high'' and ''low'' class. High-intensity species are those in which either males are commonly reported to be intolerant of one another, or male dominance hierarchies are maintained through agonistic interactions. Low-intensity species are those in which male–male agonistic contest competition is rare or absent, males are reported as relatively tolerant with normally peaceful relations, or dominance hierarchies, if present at all, are reported as difficult to detect. The potential frequency of male–male competition is likewise dichotomized into a ''high'' and ''low'' class. High-frequency competition is recognized when there is typically more than one adult male in a breeding group. Low-frequency competition is recognized when breeding groups typically contain only a single, adult male.

The four possible combinations of frequency and intensity are ranked into competition levels. Competition level 1 contains low-intensity, low-frequency species, while competition level 4 contains high-intensity, high-frequency species. The distinction between competition levels 2 and 3 is based on the intensity of male–male competition. Thus, competition level 2 contains low-intensity, high-frequency species, while competition level 3

contains high-intensity, low-frequency species. Dimorphism is predicted to increase with increasing competition level.

The male–male competition levels are designed for comparison with ratio estimates of sexual dimorphism, which by definition reflect both male and female trait variation (Plavcan & van Schaik, 1994; Plavcan *et al.*, 1995). Monogamous and polyandrous species are predicted to show low degrees of sexual dimorphism, even though male–male and female–female competition can be intense (Plavcan *et al.*, 1995). Hence, all monogamous and polyandrous species are classified as competition level 1. Further details of classification of species into competition levels, with references, are available in Plavcan (1990), Plavcan & van Schaik (1992) and Plavcan *et al.* (1995).

Any attempt to infer competition levels in extinct taxa should keep in mind that the competition levels are operational definitions of male–male competition, and mask the continuous range of variation in male–male competition seen among primates. Furthermore, the competition levels are strongly correlated with broad classifications of mating system, but the two classifications should not be equated. The most important distinction is that multimale/multifemale polygynous groups are divided into three competition levels. Thus, most competition level 3 species are single-male/multifemale groups, but multimale groups in which mating occurs during a brief mating season are also included in this group (for example, *Saimiri*). Competition level 4 species are all multimale/multifemale groups, but include species showing macaque-like male and female dominance hierarchies, and multilevel societies such as *Papio hamadryas*, *Theropithecus gelada*, and *Nasalis larvatus*, in which single male breeding groups come together to form large groups daily. Competition level 2 species include primarily male-bonded community species such as *Pan troglodytes* and *Brachyteles*, but also include *Pan paniscus*. In particular we note that, while male–male competition in both species of *Pan* is not as intense as in, for example, *Papio*, nevertheless the two species of *Pan* do differ markedly in the nature of intrasexual and intersexual relations (Nishida & Hiraiwa-Hasegawa, 1987).

Increasing sexual dimorphism in both canine tooth size and body weight is strongly correlated with increasing competition level (Plavcan, 1990, 1993 Plavcan & van Schaik, 1992, 1994, 1997; Ford, 1994). Detailed comparative analyses of the relation between competition levels and body weight dimorphism are presented in Plavcan & van Schaik (1992, 1997) and, for platyrrhines alone, Ford (1994). Detailed comparative analyses of the relation between competition levels and canine dimorphism are presented in Plavcan (1990), and Plavcan & van Schaik (1992). These analyses are not repeated here.

We present our analysis in two steps. First, we evaluate the relation between canine dimorphism, relative male canine tooth size, body weight dimorphism, and competition levels among extant species. The goal of this analysis is to evaluate how well body weight dimorphism and the different measures of canine dimorphism and relative canine size can discriminate the various competition levels among extant species. Next, we directly compare estimates of australopithecine dimorphism to dimorphism of extant species, to see whether australopithecine dimorphism is consistent with that exhibited by extant anthropoids of any particular mating system or competition level.

Results

Extant species

Dimorphism is strongly correlated among all measures (Table 3). Correlations between occlusal dimensions of the canine teeth are strongest, while those between body weight dimorphism and the canine dimensions are the weakest.

Table 3 **Pearson correlation coefficients between natural-log transformed dimorphism measures of extant anthropoids**

		Mandibular			Maxillary	
	Weight	MD	BL	Height	MD	BL
Mandibular						
MD	0·736					
BL	0·713	0·854				
Height	0·703	0·796	0·811			
Maxillary						
MD	0·679	0·917	0·857	0·826		
BL	0·733	0·903	0·864	0·731	0·906	
Height	0·628	0·812	0·808	0·912	0·890	0·7884

MD, mesiodistal; BL, buccolingual.

Table 4 presents simple descriptive statistics of means, standard deviations, ranges and sample sizes for natural-log transformed dimorphism measures of extant species within competition levels and mating system classification. Figure 1 presents box and stem diagrams illustrating these data.

Among the measures of dimorphism examined here, maxillary canine crown height shows the greatest range of dimorphism (difference between minimum and maximum values of 1·473), followed by mandibular canine crown height (1·067) and body weight (0·925). The total range of body weight dimorphism is only 63% that of maxillary canine crown height dimorphism. The total ranges of occlusal canine dimorphism are substantially narrower. Among occlusal dimensions, maxillary canine mesiodistal dimorphism shows the greatest range (0·752), which is only 51% that of the maxillary canine crown height.

Standard deviations likewise demonstrate that maxillary canine crown height dimorphism shows the greatest interspecific variation. Body weight dimorphism shows the next greatest standard deviation, followed by mandibular canine crown height dimorphism. Standard deviations of dimorphism in the occlusal dimensions are substantially lower than those of canine crown height or body weight.

In all measures of dimorphism, *average* dimorphism increases with increasing competition levels. The spread among mean dimorphism of competition levels is greatest for maxillary canine crown height dimorphism (difference between means of competition level 1 and 4 of 0·639). This is followed by mandibular canine crown height dimorphism (0·487) and body weight dimorphism (0·441). Separation between means of competition levels of occlusal dimensions is much less (ranging between 0·272 and 0·355).

The degree of overlap in ranges of dimorphism varies. However, in all measures of dimorphism, the ranges of dimorphism in competition levels 2, 3 and 4 overlap each other substantially (Figure 1). In all cases, the range of competition level 2 dimorphism falls almost entirely within the range of competition levels 3 and 4. Conversely, in all cases the magnitude of dimorphism in competition levels 3 and 4 substantially exceeds that of competition levels 1 and 2. Overlap is greatest for dimorphism of the occlusal dimensions. The best case of segregation among competition levels in dimorphism is for maxillary canine crown height dimorphism. In this case, all species of competition level 1 except for *Presbytis potenziani* fall well below the minimum value found in other competition levels. Similarly, but to a lesser degree,

Table 4 **Descriptive statistics summarizing body weight dimorphism and canine dimorphism in extant anthropoid primates within competition levels and among all extant species**

	All species	Comparison level			
		1	2	3	4
Weight dimorphism					
Mean	0·260	− 0·024	0·182	0·312	0·417
S.D.	0·235	0·068	0·080	0·215	0·168
Min.	− 0·174	− 0·174	0·077	0·000	0·115
Max.	0·802	0·086	0·307	0·802	0·751
n	85	20	8	25	32
Mandibular canine					
Mesiodistal					
Mean	0·210	0·019	0·145	0·215	0·311
S.D.	0·143	0·029	0·044	0·110	0·112
Min.	− 0·030	− 0·030	0·068	0·010	0·068
Max.	0·548	0·077	0·191	0·365	0·548
n	84	16	6	29	33
Buccolingual					
Mean	0·235	0·048	0·189	0·250	0·320
S.D.	0·142	0·051	0·030	0·117	0·114
Min.	− 0·020	− 0·020	0·148	0·039	0·077
Max.	0·582	0·122	0·215	0·582	0·565
n	84	16	6	29	33
Crown height					
Mean	0·427	0·090	0·329	0·462	0·577
S.D.	0·229	0·110	0·076	0·085	0·200
Min.	− 0·020	− 0·020	0·231	0·315	0·300
Max.	1·047	0·351	0·445	0·637	1·047
n	84	16	6	29	33
Maxillary canine					
Mesiodistal					
Mean	0·275	0·037	0·169	0·294	0·392
S.D.	0·177	0·050	0·073	0·113	0·151
Min.	− 0·020	− 0·020	0·095	0·030	0·131
Max.	0·732	0·140	0·255	0·457	0·732
n	84	16	6	29	33
Buccolingual					
Mean	0·197	0·025	0·135	0·135	0·303
S.D.	0·153	0·043	0·095	0·125	0·132
Min.	− 0·094	− 0·062	0·010	− 0·094	− 0·010
Max.	0·565	0·113	0·247	0·399	0·565
n	84	16	6	29	33
Crown height					
Mean	0·515	0·085	0·400	0·538	0·724
S.D.	0·294	0·098	0·104	0·098	0·262
Min.	− 0·062	− 0·062	0·285	0·344	0·344
Max.	1·411	0·344	0·571	0·775	1·411
n	84	16	6	29	33

the range of body weight dimorphism of competition level 1 species extends substantially lower than that of competition levels 2, 3 and 4.

Turning to measures of relative male maxillary canine size, there are distinct differences in the way that measures of relative canine size are related to competition levels (Table 5,

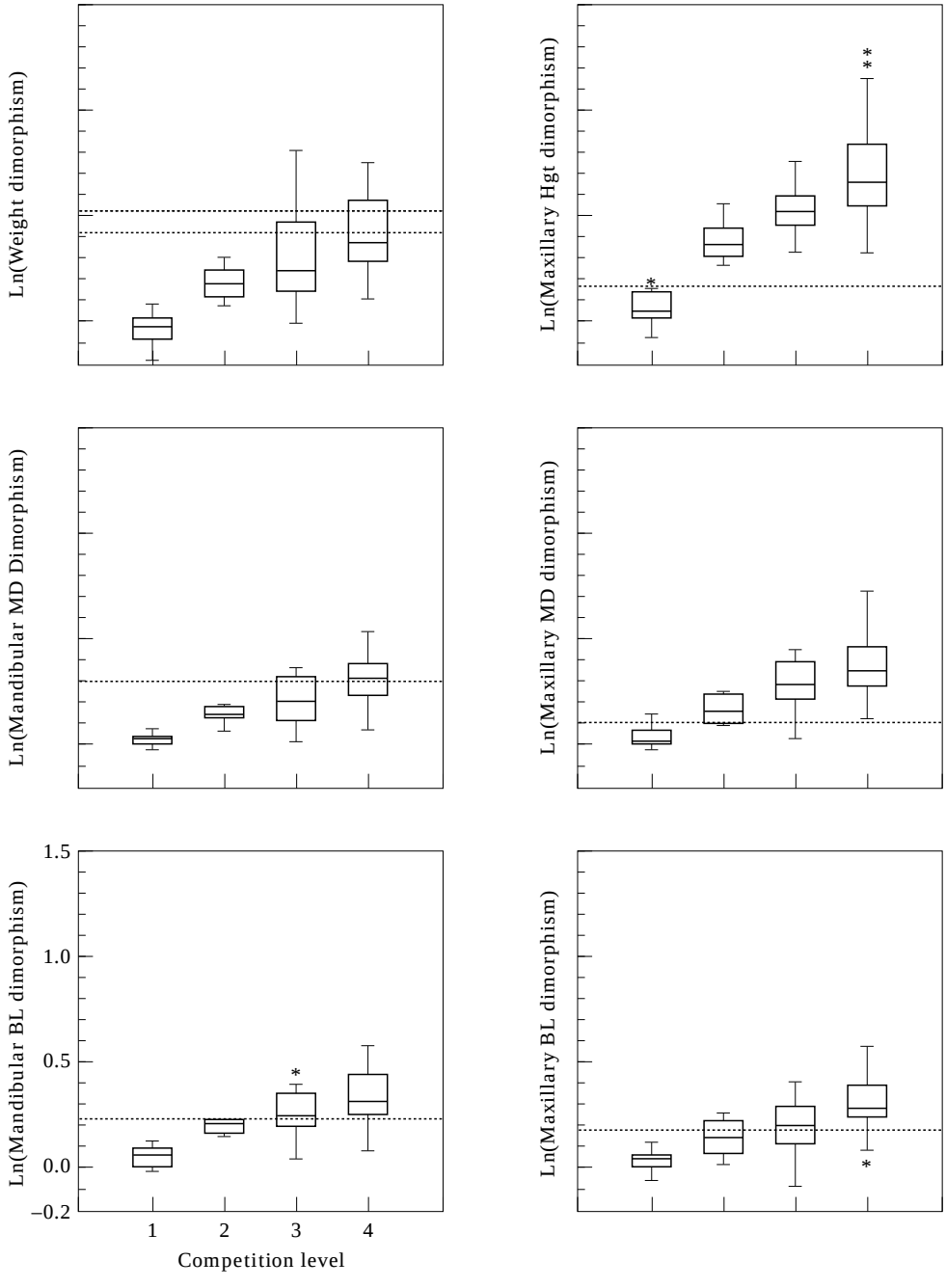


Figure 1. Box plots showing the medians and ranges of dimorphism among extant species for canine dimorphism and body weight (mandibular canine crown height dimorphism is not shown, but is similar to the figure for maxillary canine crown height). The dotted lines indicate values of estimated dimorphism for *A. afarensis*. These are shown to illustrate the comparisons between australopithecines and extant species summarized in Table 6.

Table 5 **Spearman rank correlations (r_s) between competition levels and estimates of male relative maxillary canine tooth size**

	MD	BL	Height
r_s	0·430	0·251	0·405
P	<0·001*	N.S.	<0·001

*Probabilities are based on a parametric Pearson's product-moment correlation, a close approximation recommended by Sokal & Rohlf (1995) for the Spearman rank correlation.

MD, mesiodistal; BL, buccolingual.

Figure 2). Relative male maxillary canine crown height and mesiodistal length are significantly associated with variation in competition levels. Relative male maxillary canine buccolingual breadth is not. As for measures of dimorphism, the range of relative male canine crown height and mesiodistal length for competition level 1 species extends below that of other competition levels, while the range of relative male canine crown height and mesiodistal length for competition levels 3 and 4 extends above that of the other competition levels. Unlike dimorphism estimates, the range of competition level 2 species also extends below that of competition level 3 and 4 species. Compared with estimates of dimorphism, however, the overlap among competition levels for all relative male canine crown height dimensions is more extensive.

To summarize, among the measures examined here, maxillary canine crown height dimorphism shows the greatest interspecific variation, with the greatest spread among means between competition levels. However, examining ranges of dimorphism, there is substantial overlap among competition levels 2, 3, and 4 for all measures of dimorphism. Only two definitive statements can be made about dimorphism with reference to discriminating among competition levels. First, for maxillary canine crown height dimorphism, and to a lesser degree body weight dimorphism, competition level 1 species show a range of dimorphism that extends substantially lower (that is, less dimorphism) than that of other competition levels. Second, dimorphism in all measures of many competition level 3 and 4 species substantially exceeds that of competition level 1 and 2 species. Relative male canine size is not as strongly associated with variation in competition levels as is canine dimorphism. Still, the range of relative male canine crown height and mesiodistal length of competition level 1 and 2 species extends below (that is, relative canine size is smaller) that of competition levels 3 and 4.

Hominids

Figure 1 shows values of *A. afarensis* dimorphism estimates superimposed on the box plots, to illustrate how comparisons between hominid dimorphism and extant dimorphism were evaluated. Table 6 summarizes these evaluations for each taxon, listing the competition levels whose ranges overlap each estimate of dimorphism for each australopithecine, and also the competition level showing the mean value of extant dimorphism closest in absolute value to each estimate. Figure 2 and Table 7 provide similar information for comparisons of relative maxillary canine tooth size. Figure 2 also shows comparisons between relative canine crown height of australopithecines and extant hominoids, because many studies restrict comparisons of hominid canine size to extant hominoids.

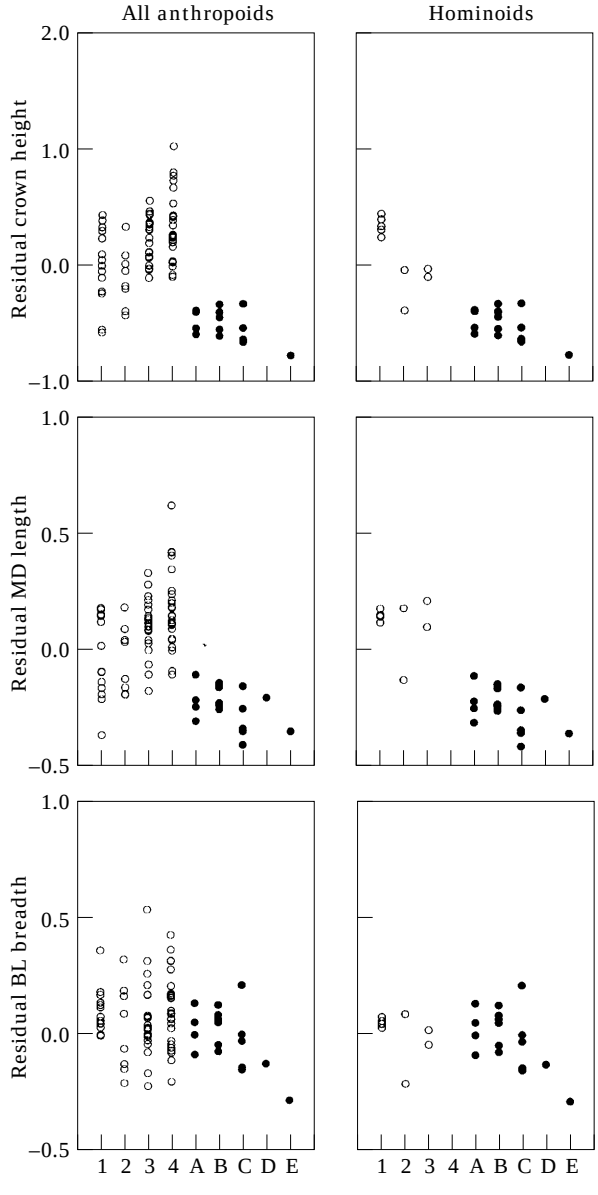


Figure 2. Plots showing the range of relative maxillary canine crown height among extant anthropoids by competition levels (open circles), and the australopithecines (filled circles). Extant species are mean values, while individual data points are shown for the australopithecines. Numbers 1–4 are competition levels. The letters represent *A. afarensis* (A), *A. africanus* (B), *A. robustus* (C), *A. boisei* (D), and *Homo* (E). The ordering of data is the same for each plot. For the hominids, individual specimen values are shown. For *A. boisei*, the values are the largest reported by Johanson *et al*. (1982). The left hand column shows all anthropoids, while the right hand column separates out the extant hominoids. Among these latter taxa, all points in competition level 1 are hylobatids, the lower point of competition level 2 is *P. paniscus*, and the higher *P. troglodytes*, the lower point of competition level 3 is *Gorilla*, and the higher *Pongo*. Table 7 summarizes the comparisons made with this figure.

Table 6 Summaries of overlapping ranges and closest means of extant primate dimorphism estimates from different competition levels to hominid dimorphism estimates

Taxon	Low weight	High weight	Mandibular			Maxillary		
			MD	BL	Height	MD	BL	Height
Australopithecus afarensis								
Range	3/4	None	3/4	(2)/3/4	—	1/2/3	2/3/4	1
Mean	4	4	4	3	—	2	2	1
Australopithecus africanus								
Range	2/3/4	3/4	2/3/4	3/4	1	1/2/3/4	2/3/4	1
Mean	3	3	2	2	1	2	3	1
Australopithecus robustus								
Range	2/3/4	2/3/4	2/3/4	2/3/4	—	1/2/3	2/3/4	1
Mean	2	2	2	2	—	2	2	1
Australopithecus boisei								
Range	3/4	3/4	1/3	—	—	2/3/4	2/3/4	
Mean	3	4	1	—	—	2	3	—
Homo sapiens								
Range	2/3/4	—	1/3	1/3	1	1/2/3	1/2/3/4	1
Mean	2	—	1	1	1	2	1	1

MD, mesiodistal; BL, buccolingual.

Table 7 Summaries of overlapping ranges and closest means of extant primate relative male maxillary canine tooth size estimates from different competition levels to hominid relative maxillary canine tooth estimates

Taxon	MD	Height
Australopithecus afarensis		
Range	1/2/3/4	1/2
Mean	1	1
Australopithecus africanus		
Range	1	1/2
Mean	1	1
Australopithecus robustus		
Range	1	1
Mean	1	1
Australopithecus boisei		
Range	1	—
Mean	1	—
Homo sapiens		
Range	1/2	1
Mean	1	1

MD, mesiodistal.

A. afarensis

Considered together, estimated dimorphism in the occlusal dimensions of the canines overlaps all four competition levels, and falls closest to the mean of either competition levels 2, 3 or 4. While dimorphism of three of the four occlusal dimensions falls within the ranges of either competition level 1 or 2, dimorphism of the mandibular canine MD dimension falls only within the ranges of competition levels 3 and 4. Only competition level 3 overlaps dimorphism

estimates of all four occlusal dimensions. Variation in estimates of dimorphism from the mean and CV techniques does not alter the results. Estimated canine crown height dimorphism clearly falls within the range of competition level 1 only. This latter observation, however, must be tempered by the fact that only four specimens were used to estimate ''dimorphism'' in *A. afarensis*. With so few specimens, there is a possibility that only one sex is represented in the sample.

More suggestive are the comparisons of relative canine size in *A. afarensis* (Figure 2, Table 7). The largest available maxillary MD dimension falls within the extant range of all four competition levels, but overlap is at the low end of competition levels 3 and 4 (exceeding only *Presbytis rubicunda* and *Presbytis melalophos* of competition level 3, and *N. larvatus* of competition level 4). The residuals of canine crown height overlap only the ranges of competition levels 1 and 2. In this latter case, the residuals overlap only *Brachyteles arachnoides* and *P. paniscus* from competition level 2. The only competition level 1 taxon with relatively shorter canines than the *A. afarensis* specimens is *Callicebus*. Examination of casts of other *A. afarensis* specimens (all showing apical wear) gave no indication that crown height was larger in other specimens. The perception that canine teeth of *A. afarensis* are relatively low-crowned by comparison to other anthropoids is pervasive in the literature on hominid dentitions (e.g. Greenfield, 1990, 1992a. Thus, unless only female specimens have been preserved in the fossil record, it appears that *A. afarensis* maxillary canines were relatively low-crowned in comparison with most extant anthropoid primates.

In contrast, dimorphism of the lowest estimate of body weight dimorphism falls only within the range of competition levels 3 and 4, but exceeds the mean value of dimorphism of competition level 4 (Figure 1, Table 6).

A. africanus

Considered together, dimorphism estimates of the occlusal canine dimensions overlap all competition levels. Three of four fall closest to the mean dimorphism of competition level 2. The mandibular BL dimorphism estimate falls closest to the mean of competition level 2, but only falls within the range of competition level 3 and 4 species. This is because the range of extant competition level 3 and 4 species actually extends below the range of competition level 2 species. Dimorphism estimates of the maxillary occlusal dimensions based on Grine's data are substantially higher than those based on data from White (1977) and Johanson *et al.* (1982). Maxillary occlusal dimorphism estimates based on the latter data set show the same pattern of overlapping ranges, but fall closest to the mean of competition level 1. Estimated mandibular and maxillary canine crown height dimorphism overlaps only competition level 1, though, as for *A. afarensis*, the small sample size opens the possibility that only males or females are present in the sample.

Estimates of relative maxillary canine crown height fall within the range of competition levels 1 and 2. As for *A. afarensis*, only *Pan* and *Callicebus* show relatively shorter canines. The two largest estimates of relative maxillary MD canine size overlap competition levels 1 and 2, and one species of competition level 3 (*P. rubicunda*). As for *A. afarensis*, the presence of only five specimens for the canine crown height measurement introduces the possibility that only males or females are present. Nevertheless, examination of other casts suggests that canine crown height did not exceed the values reported here.

The lowest estimate of body weight dimorphism overlaps competition levels 2, 3, and 4. Both estimates of weight dimorphism fall closest to the mean of competition level 3.

A. robustus

Three of four estimates of canine occlusal dimorphism and both estimates of weight dimorphism in *A. robustus* fall within the range competition levels 2, 3 and 4, but fall closest to the mean of competition level 2. The maxillary MD dimorphism estimate falls within the range of competition levels 1, 2 and 3. Using data from White (1977) and Johanson *et al.* (1982), the maxillary MD dimorphism estimate falls only within the range of competition levels 1 and 3, and falls closest to the mean competition level 1. Estimated canine crown height dimorphism overlaps only species of competition level 1. The same concerns about small sample size discussed for *A. afarensis* apply to this latter observation.

The largest estimate of relative maxillary canine crown height falls only within the range of competition levels 1 and 2. As for *A. afarensis* and *A. africanus*, the largest estimate of relative canine crown height exceeds only *Pan* and *Callicebus*. The largest estimate of relative maxillary canine MD size falls within the range of competition levels 1, 2, and 3, but only exceeds the relative male canine size of *P. rubicunda* in competition level 3. Examination of other casts of *A. robustus* canines suggests that relative canine size did not substantially exceed that of the largest values reported here.

A. boisei

Considered together, dimorphism estimates of the occlusal dimensions overlap the ranges of all four competition levels. Estimated dimorphism of each of the three occlusal dimensions falls closest to the means of competition levels 1 (mandibular MD), 2 (maxillary MD) and 3 (maxillary BL). Conversely, estimates of the maximum relative maxillary canine MD length overlap only species of competition level 1 (the value is not shown in Figure 2). Both estimates of body weight dimorphism overlap the ranges of competition levels 3 and 4, falling closest to the mean dimorphism seen in competition level 4. This latter observation should be tempered by that fact that only *two* specimens were used to estimate weight dimorphism.

Homo sapiens

Considering *H. sapiens* as if it were just another fossil hominid, canine occlusal dimorphism falls within the range of all four competition levels, and falls closest to the mean of either competition levels 1 or 2. Mandibular and maxillary canine crown height dimorphism fall only within the range of competition level 1. Body weight dimorphism falls within the range of competition levels 2, 3, and 4, but falls closest to the mean of competition level 2.

Discussion

Canine and body mass dimorphism as predictors of competition

Although measures of dimorphism in body weight and canine tooth size dimensions are highly correlated, the relation between male–male competition levels and the estimates of dimorphism is not uniform. Maxillary canine crown height shows the greatest interspecific variation in canine dimorphism, and the strongest association with male–male competition levels. Body weight dimorphism shows less interspecific variation, but nevertheless shows a strong distinction between competition level 1 and 2 species, and competition level 3 and 4 species. Dimorphism in the occlusal canine dimensions shows the least variation among extant species, and is not as strongly associated with male–male competition as dimorphism in canine crown height and body weight.

Importantly, overlap in the *range* of dimorphism is extensive among the competition levels for all measures, severely limiting what can be inferred for extinct taxa. For all measures of dimorphism examined here, very high degrees of dimorphism are associated, among extant species, only with high-intensity male–male competition. However, only for canine crown height and, to a lesser degree, body weight are very low degrees of dimorphism exclusively associated with competition level 1. Low degrees of canine occlusal dimorphism are found in taxa of all competition levels, rendering all but the highest degrees of occlusal canine dimorphism almost meaningless as evidence for the behavior of extinct taxa.

Turning to measures of relative male maxillary canine size, relative male maxillary canine crown height and mesiodistal length are significantly associated with variation in male–male competition levels, while relative male buccolingual canine breadth is not. The range of the former two measures shows extensive overlap among competition levels. Notably, competition level 1 and 2 species tend to have relatively low crowned canines.

Importantly, the extensive range of relative male maxillary canine size among competition level 1 species reflects the different ways dimorphism can be expressed, which in turn reflects variation in male and female intrasexual competition (*cf.* Plavcan, 1993; Plavcan *et al.*, 1995). All competition level 1 species are characterized by low degrees of canine dimorphism, but this only indicates that male and canine teeth are similar in size, and masks interspecific variation in relative male and female canine tooth size (Plavcan & van Schaik, 1994; Plavcan *et al.*, 1995). Thus, lack of dimorphism in the canines might reflect either large male and female canines (as in *Hylobates*), or small male and female canines (as in *Callicebus*). While species of both genera are monogamous, hylobatids aggressively defend territories from both male and female conspecifics, and show consistent ritual displays consistent with agonistic intrasexual competition in both sexes (Mitani, 1985). Conversely, male and female *Callicebus* show little aggressive intrasexual competition (Robinson *et al.*, 1987). The implication is that low levels of dimorphism can imply very different types of behavior, and emphasizes the value of even a limited comparison of relative canine size of fossils. For example, restricting an analysis of hominid canine dimorphism to extant hominoids shows that canine dimorphism is relatively slight, and perhaps most consistent with the degree of canine dimorphism seen in hylobatids. However, as seen in Figure 2, the maximum relative canine crown height of available *A. africanus* and *A. afarensis* specimens is, at most, comparable with *P. paniscus*, and is not comparable with hylobatids.

Interpreting hominid dimorphism

This analysis documents what many have noted before. Australopithecines were generally characterized by comparatively low degrees of canine tooth size dimorphism, relatively small canine teeth (in comparison with anthropoid primates), and apparently modest to strong degrees of body weight dimorphism.

Leutenegger & Shell (1987) observed that *Homo* and australopithecines share a common *pattern* of dimorphism, in that body weight dimorphism is greater than canine dimorphism. However, the pattern is not unique. Among the primates analyzed here, it is also seen in *Pongo*, *Alouatta caraya*, and *Alouatta pigra* (Table 8). That these taxa are characterized by intense male–male competition suggests that, by itself, the pattern of hominid dimorphism is not necessarily indicative of any shared behavioral pattern.

Australopithecine canine occlusal dimorphism offers apparently contradictory evidence about competition levels. On the one hand, all but four canine occlusal dimorphism estimates fall closest to the means of competition level 1 or 2 species (low-intensity male–male

Table 8 **Species from Table 1 showing a pattern of greater weight dimorphism than canine dimorphism***

Taxon	Wgt	Mand.	Max.
Alouatta caraya	1·53	1·49	1·49
Alouatta pigra	1·84	1·49	1·37
Pongo pygmaeus	2·23	1·55	1·69

*Abbreviations as follows: Wgt, weight dimorphism; Mand., mandibular canine crown height dimorphism; Max., maxillary canine crown height dimorphism. All estimates of dimorphism are simple ratios of the male divided by female value.

competition). On the other hand, all estimates of occlusal dimorphism overlap competition level 3 (high-intensity male–male competition). In *A. africanus*, *A. robustus*, and *A. boisei*, all occlusal dimorphism estimates also overlap at least competition level 1 or level 2 species (low-intensity male–male competition). Thus, in these taxa, canine occlusal dimorphism can be used to support an argument of either low-intensity male–male competition, or high-intensity male–male competition. On the other hand, *A. afarensis* mandibular MD dimorphism overlaps only competition level 3 and 4 species. This latter observation would suggest that a reconstruction of high-intensity male–male competition is most parsimonious for occlusal dimorphism in this species.

We suggest that australopithecine canine occlusal dimorphism actually offers little information about either male–male competition or mating systems. Recent analyses of canine dimorphism and relative canine tooth size in extant primates (Greenfield, 1990,1992*a,b*; Greenfield & Washburn, 1991; Plavcan, 1993; Plavcan & van Schaik, 1992, 1994; Plavcan *et al.*, 1995) suggest that relative canine crown height, not occlusal canine size, is the most important aspect of the tooth in its function as a weapon. Canine occlusal dimorphism is only related to canine crown height dimorphism through the correlation between canine occlusal size and crown height. Notably, however, taxonomic groups differ substantially in the scaling of canine height against occlusal dimensions, and such shape differences in the canine teeth are not associated with variation in competition levels (Plavcan, 1993). Therefore, evidence from crown height dimorphism and relative canine crown height should be strongly preferred over any evidence from occlusal dimorphism.

Estimates of maxillary canine crown height dimorphism of *A. afarensis*, *A. africanus*, and *A. robustus* are consistent only with competition level 1 species. The small sample sizes of crown height data for each of these species means that the dimorphism estimates should be approached with caution. However, all three hominid comparisons give low estimates of crown height dimorphism, suggesting that crown height dimorphism was indeed relatively low in these taxa.

Relatively small canine tooth size is considered a definitive hominid trait (White *et al.*, 1993). For *A. afarensis*, *A. africanus*, and *A. robustus*, the maximum maxillary canine MD length overlaps species of competition levels 1 and 2, but also overlaps the low end of the range of competition level 3 (and, for *A. afarensis*, competition level 4). Among extant competition level 3 and 4 species, only *N. larvatus*, *Presbytis rubicunda*, and *P. melalophos* show relatively smaller male MD maxillary canine dimensions. However, each of these species also shows relatively tall canine crown heights. In contrast, measures of relative maxillary canine crown height in *A. afarensis*, *A. africanus* and *A. robustus* overlap only species of competition levels 1 and 2 (no crown height

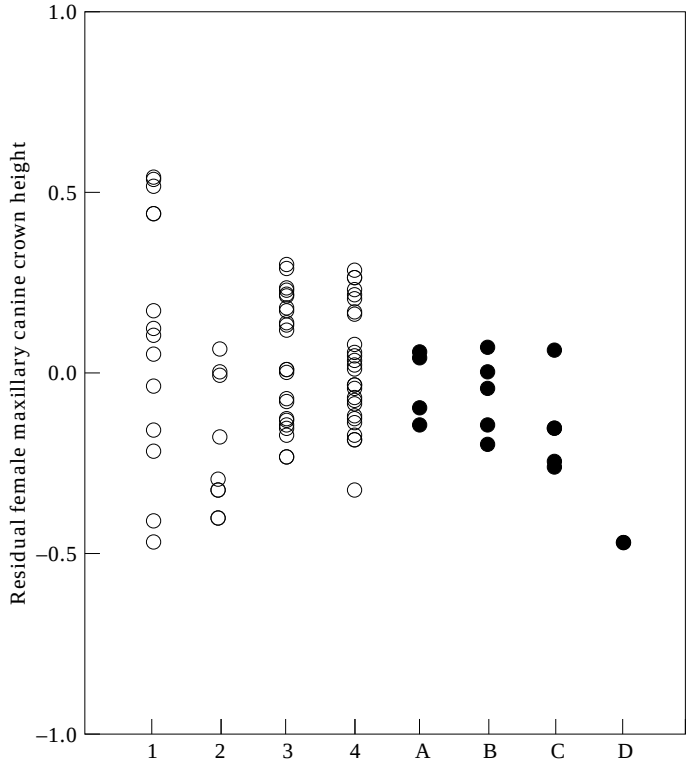


Figure 3. Plot comparing residual maxillary canine crown height of female anthropoids to that of australopithecines and *Homo*. Residual values are from a least squares regression between ln-transformed crown height and ln-transformed female body weight. For the australopithecines, McHenry's (1992) lowest estimate of female body weight was used. For the australopithecines, individual data points are shown. Numbers 1–4 are competition levels. Hominids are *A. afarensis* (A), *A. africanus* (B), *A. robustus* (C), and *Homo* (D).

data was available for *A. boisei*). The closest extant analogies for relative canine crown height of these three taxa are *P. paniscus*, *B. arachnoides*, and *Callicebus*. We used the lowest estimates of male body weight reported by McHenry (1992) to calculate relative male canine crown height. Substituting the larger body weight estimates, relative canine size in all hominids is even smaller. Even substituting the smallest estimates of female body weight reported by McHenry (thus maximizing the estimates of australopithecine relative maxillary canine crown height), hominid canine crown height falls comfortably in the size range characteristic of *female* anthropoid primates (Figure 3). Considered together, these results suggest that, unless only female hominid canines are preserved in the fossil record, relative male canine size in all four australopithecine taxa is comparable only with competition level 1 or 2 species.

In both *A. africanus* and *A. robustus*, estimated weight dimorphism overlaps competition levels 2, 3 and 4. In these cases, overlap of weight dimorphism estimates with competition level 2 presents a more consistent picture with the canine dimorphism data because both competition levels 1 and 2 include taxa that are characterized by low-intensity male–male competition. Interestingly, humans show a similar pattern of overlap, with canine dimorphism and relative maxillary canine size consistent with competition level 1 or 2, and weight dimorphism falling

within the range of competition levels 2, 3, and 4, but closest to the mean of competition level 2. It thus seems reasonable to conclude that, for *A. africanus* and *A. robustus*, body weight dimorphism, canine dimorphism, and relative canine size are at least *compatible* with a reconstruction of low-intensity male–male competition. Obviously, the degree of overlap of the various dimorphism estimates with all competition levels precludes any definitive statement.

In contrast, even the lowest body weight dimorphism estimates provided by McHenry (1992) in two of the four australopithecines (*A. afarensis* and *A. boisei*) are compatible only with competition level 3 or 4 species. In both cases, this conflicts with the canine crown height data. Competition level 3 and 4 species are characterized by intense male–male competition, usually for access to mates in a strongly polygynous mating system. Competition level 1 and 2 species are characterized either by relatively tolerant male–male relations, at least within groups, or by a monogamous or polyandrous mating system.

The apparently paradoxical pattern of weight and canine dimorphism in *A. afarensis* and *A. boisei* presents a difficult problem for using dimorphism to infer behavior. Assuming that the taxonomic assignments are correct, and that the dimorphism and relative canine tooth size estimates are reasonably accurate, dimorphism of one character (canines or inferred body weight) can only be used as evidence of behavior if dimorphism in the other character is assumed to reflect some unusual mechanism operating in hominids. In the discussion that follows, we systematically evaluate a variety of alternative explanations that may help account for the observed pattern of dimorphism and relative canine size in these two taxa.

To begin, the apparent paradox of dimorphism in *A. afarensis* and *A. boisei* can potentially be resolved by arguing that relative canine tooth size and canine dimorphism offer more ''reliable'' evidence of male–male competition than body weight dimorphism. After all, maxillary canine crown height dimorphism is most strongly associated with variation in competition levels, while variation in relative canine size offers more refined evidence of male–male competition (Plavcan *et al.*, 1995). Furthermore, the large canine teeth of most male anthropoids seems to function primarily as weaponry for intraspecific combat (Greenfield, 1992b; Plavcan & van Schaik, 1992; Plavcan *et al.*, 1995), whereas male and female body size are probably under a variety of conflicting selective pressures, of which male–male competition is only one (Ford, 1994; Harvey *et al.*, 1978; Leigh & Shea, 1995; Leutenegger & Kelley, 1977; Martin *et al.*, 1994; Mitani *et al.*, 1995). Nevertheless, our analysis shows that *both* body weight dimorphism and canine crown height dimorphism are strongly associated with male–male competition levels. Empirically, canine crown height dimorphism offers only slightly better ''resolution'' than body weight dimorphism among competition levels. Therefore, the paradox of canine and weight dimorphism in *A. afarensis* and *A. boisei* cannot be resolved by simply arguing that body weight dimorphism gives a ''bad signal''.

If canine dimorphism and relative canine tooth size are indicative of behavior, the relatively low crowned maxillary canines of all australopithecines imply a social system either characterized by low levels of intrasexual competition (as for example, in *P. troglodytes*, *P. paniscus*, *Brachyteles*, *Callicebus*), or where male and female intrasexual competition results in similar development of weaponry in both sexes (as in *Hylobates*). The low crowned canines of all australopithecines suggest that the former would be the case. The implication of low levels of male–male competition is consistent with the suggestion of a number of studies that australopithecines were characterized by a social system similar to that of chimpanzees or bonobos (e.g. Wrangham, 1987). However, on the basis of canine dimorphism and relative canine tooth size alone, one can only say that the canines were not important as weapons for

intrasexual competition, probably in association with a lack of intense male–male competition (Plavcan *et al.*, 1995). Beyond this, canine dimorphism and relative canine size say little about any particular mating system.

Assuming that relative canine tooth size is indicative of behavior, the unusually high degrees of weight dimorphism in *A. afarensis* and *A. boisei* are difficult to explain. We can imagine six potential explanations: (1) sampling error leads to overestimation of weight dimorphism, (2) the range of extant competition level 2 taxa is undersampled, (3) weight dimorphism and canine dimorphism respond differently to a relaxation of selection for larger male body weight and weaponry, (4) the presence of two species in the data leads to an overestimate of weight dimorphism, (5) there was selection for large male body size for predator defense, and (6) there was selection favoring a reduction in female body weight. We discussed each of these briefly in turn.

Sampling error. The sample sizes that McHenry (1992) uses to estimate body weight are relatively small (McHenry's estimate of *A. boisei* dimorphism is based on only *two* specimens), opening up the possibility that weight dimorphism is overestimated (or, of course, underestimated) in any or all of the australopithecines. In fact, the methods used to estimate dimorphism in all australopithecines overestimate dimorphism when the true sample dimorphism is relatively low. However, other recent analyses of australopithecine cranial and skeletal elements suggest that size dimorphism was substantial, especially in *A. afarensis* and *A. boisei* (Lockwood *et al.*, 1996; Hartwig-Scherer, 1993; Richmond & Jungers, 1995; Aiello & Wood, 1994). Still, we emphasize that the uncertainty in weight dimorphism estimates presents a major problem for interpreting the implications of australopithecine dimorphism, and we must be open to the possibility that further discoveries could alter perceived degrees of size dimorphism in any of these species. Until this happens, however, we are forced to work with the hypothesis that *A. afarensis* at least was characterized by moderate to strong body weight dimorphism.

Undersampled extant ranges. McHenry (1994*b*) argues that the degree of body size dimorphism in early hominids is consistent with competition level 2 species such as *Pan* and *Ateles*. In other publications, McHenry likewise states that the size dimorphism of *A. afarensis* is slightly greater than that of *Pan*, and well below that of *Pongo* or *Gorilla*. In part, the difference between McHenry's comparison and ours are the estimates of weight dimorphism of *Pan* and *Gorilla*. McHenry uses an estimate of 1·44 for *P. troglodytes*, and 2·03 for *Gorilla*. The value for *Pan* is higher than weight dimorphism estimates reported by other sources, while that of *Gorilla* is substantially higher (Table 9). Still, body weight dimorphism estimates for many extant species are notoriously variable (Plavcan & van Schaik, 1997), and it is possible that the range of competition level 2 taxa could extend somewhat higher. Regardless, the lowest estimate of weight dimorphism for *A. afarensis* exceeds *any* currently available estimate of weight dimorphism of any competition level 2 species represented in our data set.

Another possibility is that the range of dimorphism of extant competition level 2 species is undersampled. Only eight taxa are represented in this sample, and there is a possibility that a complete sampling of anthropoid primates would extend the range of dimorphism of competition level 2. Until this is done, however, such a possibility remains only speculative. Our sample of body weight data includes at least one species of all anthropoid genera except *Allenopithecus*, *Simias*, *Callimico*, *Rhinopithecus*, and *Pygathrix*. Plavcan (1990) and Plavcan & van Schaik (1992) classify *Simias concolor* and *Pygathrix nemaeus* as level 3, and *Rhinopithecus roxellana* as

Table 9 Published body mass (kg) and dimorphism estimates of gorillas, chimpanzees, and bonobos*

Male	Female	Dimorphism	Source
Gorilla gorilla			
159·20 (7)	97·70 (8)	**1·63**	Jungers (1985)
160·00	93·00	1·72	Harvey *et al.* (1987)
160·00	95·00	1·68	Stewart & Harcourt (1987)
Pan paniscus			
45·00 (10)	33·20 (10)	**1·36**	Jungers (1985)
54·50 (2)	37·00 (1)	1·47	Coolidge & Shea (1982)
38·30 (8)	32·20 (5)	1·19	Corruccino & McHenry (1979)
39·80 (8)	31·20 (8)	1·28	Cramer & Zihlman (1978)
47·80 (5)	33·10 (7)	1·44	McHenry (1991)
45·50 (4)	33·40 (6)	1·36	Morbeck & Zihlman (1989)
Pan troglodytes schweinfurthi			
43·00 (15)	33·20 (10)	**1·30**	Jungers & Susman (1984)
42·30 (11)	30·00 (7)	1·41	Pusey (1978) cited in Uehara & Nishida (1987)
39·50 (9)	29·80 (6)	1·32	Wrangham & Smuts (1980) cited in Uehara & Nishida (1987)
42·80 (6)	35·20 (8)	1·19	Uehara & Nishida (1987)
Pan troglodytes troglodytes			
60·00 (20)	47·40 (20)	1·26	Jungers (1985)
Pan troglodytes (no specified subspecies)			
54·20 (6)	39·70 (9)	1·36	McHenry (1991)
56·69 (1)	43·99 (1)	1·29	Crile & Quiring (1940)
39·50	29·80	1·33	Rodman (1984)
41·60	31·10	1·34	Harvey *et al.* (1987)
49·00	41·00	1·20	Gaulin & Sailer (1984)

*Numbers in parentheses are the reported sample sizes. Values of dimorphism in bold type are those used in this analysis.

level 4. Comparatively little is known about the behavior of *Allenopithecus* or *Callimico*. Therefore, we regard it as possible, but unlikely, that the range of dimorphism of competition level 2 species will be extended upward.

Differential response to relaxed selection. It is possible that a reduction in selection for relatively large male canines or body weight may result in a relative reduction in the size of one character, but not the other. However, the dissociation in magnitude of dimorphism for *A. afarensis* and *A. boisei* is not satisfactorily explained in this way. There are a number of anthropoids characterized by intense male–male competition that show low degrees of weight dimorphism coupled with strong canine dimorphism (e.g. several *Presbytis*), but no species characterized by low-intensity male–male competition shows such high degrees of weight dimorphism coupled with low degrees of canine dimorphism. While it is not difficult to imagine selective constraints on an increase in male body size (*cf.* Demment, 1983), it seems unlikely that selection would not favor a reduction in male body size in the absence of sexual selection, especially given the overall adaptive importance of body size.

Multiple taxa. Another possible explanation for strong body size dimorphism in australopithecines is that there are two or more species being sampled, with the mixture of individuals lending an impression of extreme variability in size. It has long been noted that metric and

morphological variation in *A. afarensis* is high (Falk & Conroy, 1983; Ferguson, 1983, 1984; Johanson *et al.*, 1978; Johanson & White, 1979; Leakey, 1981; Olsen, 1981, 1985; Schmid, 1989; Senut, 1986; Senut & Tardieu, 1985; Zihlman, 1985). Recent discoveries from Hadar (Kimbel *et al.*, 1994) and Maka (White *et al.*, 1993) reportedly extend the range of variation in cranial, dental, and postcranial features, re-igniting the debate about the specific composition of *A. afarensis*. A recent analysis of the pelvis of AL 288-1 suggests that this specimen may have been a male (Hausler & Schmid, 1995). If true, this would suggest that two taxa are present in *A. afarensis*. The presence of multiple taxa has been used to explain high levels of size dimorphism in *Homo habilis* (Lieberman *et al.*, 1988; McHenry, 1992; Wood, 1993).

For *A. afarensis*, supporters of the single species hypothesis appropriately point out that metric variation in cranial, dental and postcranial characters for all hominids does not clearly exceed that seen in extant, dimorphic species, and that there are no clear, discrete characters that allow the sorting of specimens into two distinct taxa (Kimbel & White, 1988). While it is true that even the highest estimates of dimorphism in *A. afarensis* do not exceed the level of dimorphism seen in the most dimorphic primates, our analysis suggests that the relative magnitude of the pattern of dimorphism in *A. afarensis* is very unusual in comparison with all extant primates and even other hominids. The lack of a compelling explanation for how the pattern of *A. afarensis* dimorphism could evolve is suggestive. Still, the finding of a similar pattern of dimorphism for all hominids, incuding *Homo*, as well as the observation that the problem is not unique to *A. afarensis* suggests that, unless *A. afarensis* and *A. boisei* are both composed of multiple taxa (or that size dimorphism in *A. boisei* is subtantially overestimated), this hypothesis does not satisfactorily explain the unusual pattern of dimorphism in either taxon.

Predator defense. Predator defense has long been hypothesized to explain higher degrees of weight and canine dimorphism in terrestrial anthropoids (e.g. Clutton-Brock *et al.*, 1977). This model assumes that males alone defend troops (or are susceptible to predators during transfer), and that increased body size offers a survival advantage to males that defend against predators. Unfortunately, the hypothesis does not explain why large canines are not also useful to male hominids in predation defense, in addition to large body size. It could be posited that canines are no longer useful because they have been supplanted by hand-held weapons, but this explanation is problematic (see below), and does not explain why body size dimorphism is maintained. It also does not explain why only two of the four australopithecines show unusually high degrees of size dimorphism because all should have been susceptible to predators.

Reduction in female body size. A reduction in female body size might explain the high degree of body size dimorphism in *A. afarensis* (Leigh, pers. comm.). Leigh & Shea (1995) demonstrate that body weight dimorphism in *P. troglodytes* may be reduced by an increase in female body size associated with later female maturation relative to other African apes. This delay in female maturation is hypothesized to be associated with exploitation of more scattered resources, and hence greater female–female resource competition. Reversing this logic, an increase in body weight dimorphism in *A. afarensis* could result from a shift to a more uniform resource distribution, leading to less female–female contest competition for resources, earlier female maturation, and hence reduced female body size. Such an analysis is consistent with McHenry's (1992) suggestion that reduced dimorphism in latter hominids is associated with an increase in female body size, rather than a reduction in male body size. However, the analogy

to extant hominoids must be stretched to achieve the *A. afarensis* degree of size dimorphism (Leigh, pers. comm.). Furthermore, the shift in resource competition referred to in this model seems analogous to the distinction between scramble and contest competition for resources among female primates. If this latter analogy holds, the hypothesis receives little support from comparative evidence. Among primates, scramblers tend to be folivores or insectivores, which, with only a few exceptions, tend to show relatively low degrees of body weight dimorphism (Plavcan & van Schaik, 1997). While female scramble competition may help explain the exaggerated dimorphism seen in *Gorilla*, nevertheless the relatively large male body size of *Gorilla* is also consistent with the hypothesis that sexual selection associated with intense male–male competition also contributes to the exaggerated dimorphism of this species. In fact, outside of the colobines (which show *reduced* body weight dimorphism), scramblers (such as *Alouatta* spp. and *Brachyteles*) show body weight dimorphism little different from other species of similar competition levels. The large degree of body size dimorphism should thus imply polygyny in *A. afarensis* whether they were scramblers or not. Nevertheless, we cannot exclude this hypothesis as a potential explanation for relatively strong size dimorphism in *A. afarensis*.

The above discussion assumes that relative canine tooth size is a better indicator of behavior than body size dimorphism. The alternative is that early hominid body size dimorphism (for at least *A. afarensis* and *A. boisei*) reflects polygynous mating systems with intense male–male competition. If so, then the low-crowned canines must be explained.

As for size dimorphism, there is the possibility that sampling error is responsible for the perception of low degrees of canine crown height dimorphism and relatively low-crowned canines. After all, all estimates of occlusal canine dimorphism are consistent with competition level 3 species. As discussed above, we believe that current models for the evolution of canine dimorphism strongly suggest that canine crown height is the target of selection for the development of weaponry in anthropoid primates. Furthermore, this hypothesis would require that only female canines are preserved in the fossil record. We regard this as extremely unlikely.

A number of hypotheses for canine reduction in humans have been proposed which do not directly link canine tooth size to competition. Greenfield (1992*a*) recently reviewed these models, concluding that none receive strong support. To summarize his review, the idea that canines are reduced because of tooth crowding (in association with reduction of the jaws) is not supported by any other primate that shows reduction in jaw size. Likewise, the hypothesis that canine reduction is associated with early premaxillary/maxillary sutural fusion is not supported by comparative evidence. The probable mutation effect has long been recognized as fundamentally untestable, but even so requires that the canines are not selected for development as weapons. Masticatory inhibition in association with dietary changes is inconsistent with comparative evidence, and is, to our knowledge, not supported by *in vitro* studies of mastication. Niche differentiation is unsupported by field data, and lacks an underlying mechanism outside of the context of intrasexual competition. In addition to Greenfield's evaluations, we also note that Greenfield's (1992*a*) hypothesis that canines are selected for use as incisors is functionally improbable (Plavcan & Kelley, 1996). Lande's (1980) model of correlated response does not apply because it can only (potentially) explain why female canines would be large—not why male canines would be secondarily reduced in size. Finally, phylogenetic constraints cannot be invoked because canine reduction has clearly occurred as a derived condition relative to other hominoids (White *et al.*, 1994). Current models for the evolution of canine size in primates only suggest that when the canines are not used as weapons, some factor (as yet unknown) acts to quickly reduce canine size.

The oldest hypothesis explaining the evolution of the human pattern of sexual dimorphism is that of Darwin (1871), who suggested that the use of weapons replaced the use of canine teeth in intrasexual combat, thereby resulting in a lack of selection for the development of weaponry and a reduction in canine size in human males, even though the mating system was characterized by high degrees of intermale competition. The weapons replacement hypothesis is supported by the observation that modern humans have small canines and use hand-held weapons. However, given that no other primate reduces its canines in association with weapons replacement, the hypothesis cannot be tested with a comparative method.

Nonetheless, several lines of criticism can be directed to the weapons-replacement hypothesis. First, even though hand-held tools might help in fights, it seems that large canines, like knives, still would be very useful for close physical fighting. Second, if hand-held weapons were so effective that they eliminated selection for large canines, it seems unusual that weapons did not at least moderate the advantages of larger male body size. Wolpoff (1976) argues that larger size would help an individual more effectively wield weapons, but we could also argue that the advantages of larger size would be counteracted by the development of fighting skills and agility in implementing weapons (Kappeler, 1990). Third, canine reduction is not associated with the appearance of stone tools, which show up more than a million years after the appearance of *A. afarensis*. Finally, even if the weapons replacement hypothesis helped to explain canine reduction in hominids, it still does not resolve the paradox of dimorphism in *A. afarensis*. For example, assume that the common ancestor of *Pan* and the hominids showed a degree of canine dimorphism similar to *Pan* [a reasonable hypothesis given the canine morphology of *A. ramidus* (White *et al.*, 1994)]. Few would argue that canine dimorphism in *Pan* is reduced relative to other hominoids through weapons-replacement in intermale competition. It follows that the degree of body weight dimorphism in *A. afarensis* would still be inconsistent with the degree of canine dimorphism by comparison to extant competition level 2 species. Thus, the weapons replacement hypothesis is not particularly satisfying in resolving the problem of dimorphism in *A. afarensis*.

Finally, in light of this discussion, it is tempting to simply assert that unique processes have led to the evolution of a unique pattern of dimorphism in the hominid lineage, and therefore that extant anthropoids cannot be used as analogies for evaluating the implications of hominid dimorphism. Certainly, if the weight and canine dimorphism estimates evaluated here are accurate, and the taxonomic assignments are valid, at least some hominids show a unique pattern and degree of dimorphism among primates. However, there is no a priori reason to believe that evolutionary processes at work in other primates do not also work in hominids. Even if there is no direct analog for the pattern and degree of dimorphism seen in some hominids, we still believe that extant models, with progressive refinement, offer the best opportunity for understanding the processes that governed the evolution of dimorphism in hominids. Our analysis does not necessarily imply the action of unique process in hominids. Rather, it underscores the necessity of further work to understanding the evolution of dimorphism in general, in order to understand the evolution of dimorphism in hominids.

Conclusions

Sexual dimorphism in canine tooth size and body weight offers a limited opportunity to infer male–male competition and mating system in extinct species. For extant species, canine crown height dimorphism is the best discriminator among competition levels, while dimorphism of

the canine occlusal dimensions is the poorest. Inferred relative canine crown height offers additional important information about the use of canines as weapons.

Estimates of australopithecine canine dimorphism, size dimorphism, and relative canine size do not offer clear, unambiguous evidence about the mating systems of these animals. Estimates of canine occlusal dimorphism apparently offer little information about mating system. Considered alone, canine crown height dimorphism and relative maxillary canine crown height suggests that australopithecines were characterized by reduced selection for the development of weaponry in males, implying a mating system characterized by low levels of male–male competition. McHenry's (1992) estimates of body size dimorphism in *A. robustus* and *A. africanus* can be reconciled with this evaluation. Conversely, McHenry's (1992) estimates of weight dimorphism in *A. afarensis* and *A. boisei* are consistent only with anthropoids characterized by intense male–male competition and strongly polygynous mating systems. No current model satisfactorily reconciles the behavioral implications of canine and weight dimorphism in these two taxa. Therefore, assertions that dimorphism in either canine size or body weight are indicative of a particular mating system in either of these taxa cannot be supported by an extant analogy.

The only certain conclusion of this analysis is that the behavioral implications of dimorphism in the australopithecines are *not* certain, in spite of recent advances in understanding the behavioral and ecological correlates of dimorphism in extant primates. There has been a tendency in the literature to focus on dimorphism in a single character as indicative of behavior, and to explain away inconsistencies in the pattern of dimorphism with post hoc scenarios that have little or no empirical support from comparative models. The apparent paradox of dimorphism presented by *A. afarensis* and *A. boisei* suggests to us that considerable work remains in order to understand the behavioral implications and evolution of sexual dimorphism in australopithecines.

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References

Aiello, L. C. & Wood, B. A. (1994). Cranial variables as predictors of hominine body mass. *Am. J. phys. Anthrop.* **95,** 409–426.

Cheverud, J. M., Dow, M. M. & Leutenegger, W. (1985). The quantitative assessment of phylogenetic constraints in comparative analysis: sexual dimorphism in body weight among primates. *Evolution* **38,** 1335–1351.

Clutton-Brock, T. H., Harvey, P. H. & Rudder, B. (1977). Sexual dimorphism, socionomic sex ratio and body weight in primates. *Nature* **269,** 191–195.

Cole, T. M. & Smith, F. H. (1987). An odontometric assessment of variability in *Australopithecus afarensis. J. hum. Evol.* **3,** 221–234.

Coolidge, H. J. & Shea, B. T. (1982). External body dimensions of *Pan paniscus* and *Pan troglodytes* chimpanzees. *Primates* **23,** 245–251.

Cope, D. A. (1993). Measures of dental variation as indicators of multiple taxa in samples of sympatric *Cercopithecus* species. In (W. L. Kimbel & L. B. Martin, Eds) *Species, Species Concepts, and Primate Evolution*, pp. 211–237. New York: Plenum Press.

Corruccini, R. S. & McHenry, H. M. (1979). Morphological affinities of *Pan paniscus. Science* **204,** 1341–1342.

Cramer, D. L. & Zihlman, A. L. (1978). Sexual dimorphism in the pygmy chimpanzee, *Pan paniscus*. In (D. J. Chivers & K. A. Joysey, Eds) *Recent Advances in Primatology, Vol. 3. Evolution*, pp. 487–490. London: Academic Press.
Crile, G. & Quiring, D. P. (1940). A record of the body weight and certain organ and gland weights of 3690 animals. *Ohio J. Sci.* **40,** 219–259.
Darwin, C. (1871). *The Descent of Man, and Selection in Relation to Sex*. London: J. Murray.
Demment, M. W. (1983). Feeding ecology and the evolution of body size of baboons. *Afr. J. Ecol.* **21,** 219–233.
Ely, J. & Kurland, J. A. (1989). Spatial autocorrelation, phylogenetic constraints, and the causes of sexual dimorphism in primates. *Int. J. Primatol.* **10,** 151–171.
Falk, D. & Conroy, C. G. (1983). The cranial venous sinus system in *Australopithecus afarensis. Nature* **306,** 779–781.
Ferguson, W. W. (1983). An alternative interpretation of *Australopithecus afarensis* fossil material. *Primates* **24,** 397–409.
Ferguson, W. W. (1984). Revision of fossil hominid jaws from the Plio/Pleistocene of Hadar, in Ethiopia, including a new species of the genus *Homo* (Hominoidea: Hominidae). *Primates* **25,** 519–529.
Foley, R. A. & Lee, P. C. (1989). Finite social space, evolutionary pathways, and reconstructing hominid behavior. *Science* **243,** 901–906.
Ford, S. M. (1994). Evolution of sexual dimorphism in body weight in platyrrhines. *Am. J. Primatol.* **34,** 221–244.
Gaulin, S. J. C. & Sailer, L. D. (1984). Sexual dimorphism in weight among the primates: the relative impact of allometry and sexual selection. *Int. J. Primatol.* **5,** 515–535.
Gingerich, P. D. & Schoeninger (1979). Patterns of tooth size variability in the dentition of primates. *Am. J. phys. Anthrop.* **51,** 457–466.
Godfrey, L. R., Lyon, S. K. & Sutherland, M. R. (1993). Sexual dimorphism in large-bodied primates: the case of the subfossil lemurs. *Am. J. phys. Anthrop.* **90,** 315–334.
Greenfield, L. O. (1990). Canine reduction in early man: a critique of three mechanical models. *Hum. Evol.* **5,** 213–226.
Greenfield, L. O. (1992*a*). Origin of the human canine: a new solution to an old enigma. *Yearb. Phys. Anthropol.* **35,** 153–185.
Greenfield, L. O. (1992*b*). Relative canine size, behavior, and diet in male ceboids. *J. hum. Evol.* **23,** 469–480.
Greenfield, L. O. & Washburn, A. (1991). Polymorphic aspects of male anthropoid canines. *Am. J. phys. Anthrop.* **84,** 17–34.
Grine, F. E. (1989). New hominid fossils from the Swartkrans Formation (1979–1986 excavations): craniodental specimens. *Am. J. phys. Anthrop.* **79,** 409–449.
Grine, F. E. (1993). Description and preliminary analysis of new hominid craniodental fossils from the Swartkrans Formation. In (C. K. Brain, Ed.) *Swartkrans: A Cave's Chronicle of Early Man*, pp. 75–116. Pretoria, Transvaal Museum.
Hartwig-Scherer, S. (1993). Body weight prediction in early fossil hominids: towards a taxon-''independent'' approach. *Am. J. phys. Anthrop.* **92,** 17–36.
Harvey, P. H., Kavanagh, M. & Clutton-Brock, T. H. (1978). Sexual dimorphism in primate teeth. *J. Zool. Lond.* **186,** 474–485.
Harvey, P. H., Martin, R. D. & Clutton-Brock, T. H. (1987). Life histories in comparative perspective. In (B. B. Smuts, D. L. Cheney, R. M. Seyfarth, R. W. Wrangham & T. T. Struhsaker, Eds) *Primate Societies*, pp. 181–196. Chicago, University of Chicago Press.
Hausler, M. & Schmid, P. (1995). Comparison of the pelves of Sts 14 and AL 288-1: Implications for birth and sexual dimorphism in australopithecines. *J. hum. Evol.* **29,** 363–383.
Johanson, D. C. & White, T. D. (1979). A systematic assessment of early African hominids. *Science* **202,** 321–330.
Johanson, D. C., White, T. D. & Coppens, Y. (1978). A new species of the genus *Australopithecus* (Primates: Hominidae) from the Pliocene of Eastern Africa. *Kirtlandia* **28,** 2–14.
Johanson, D. C., Taieb, M. & Coppens, Y. (1982). Pliocene hominids from Hadar formation, Ethiopia (1973–1977): stratigraphic, chronologic, and paleoenvironmental contexts, with notes on hominid morphology and systematics. *Am. J. phys. Anthrop.* **57,** 373–402.
Jolly, C. (1970). The seed eaters: a new model of hominid differentiation based on a baboon analogy. *Man* **5,** 5–26.
Jungers, W. L. (1985). Body size and scaling of limb proportions in primates. In (W. L. Jungers, Ed.) *Size and Scaling in Primate Biology*, pp. 345–381. New York: Plenum Press.
Jungers, W. L. (1988). New estimates of body size in australopithecines. In (F. E. Grine, Ed.) *Evolutionary History of the ''Robust'' Australopithecines*, pp. 115–125. New York: Aldine de Gruyter.
Jungers, W. L. & Susman, R. L. (1984). Body size and skeletal allometry in African apes. In (R. L. Susman, Ed.) *The Pygmy Chimpanzee: Evolutionary Biology and Behavior*, pp. 131–177. New York: Plenum Press.
Kappeler, P. M. (1990). The evolution of sexual size dimorphism in prosimian primates. *Am. J. Primatol.* **21,** 201–214.
Kay, R. F. (1982). *Sivapithecus simonsi*, a new species of Miocene hominoid, with comments on the phylogenetic status of Ramapithecinae. *Int. J. Primatol.* **3,** 113–173.
Kay, R. F., Plavcan, J. M., Glander, K. E. & Wright, P. C. (1988). Sexual selection and canine dimorphism in New World monkeys. *Am. J. phys. Anthrop.* **77,** 385–397.
Kelley, J. (1989). Behavioral implications of sexual dimorphism in *Australopithecus afarensis. Am. J. phys. Anthrop.* **78,** 251.
Kimbel, W. L. & White, T. D. (1988). Variation, sexual dimorphism and the taxonomy of Australopithecus. In (F. E. Grine, Ed.) *Evolutionary History of the ''Robust'' Australopithecines*, pp. 175–192. New York: Aldine de Gruyter.

Kimbel, W. H,., Johanson, D. C. & Rak, Y. (1994). The first skull and other new discoveries of *Australopithecus afarensis* at Hadar, Ethiopia. *Nature* **368,** 449–451.
Lande, R. (1980). Sexual dimorphism, sexual selection and adaptation in polygenic characteristics. *Evolution* **34,** 292–307.
Leakey, M. D. (1981). Tracks and tools. *Phil. Trans. R. Soc. Lond. Ser. B* **292,** 95–102.
Leigh, S. R. & Shea, B. T. (1995). Ontogeny and the evolution of body size dimorphism in apes. *Am. J. Primatol.* **36,** 37–60.
Leutenegger, W. (1982). Scaling of sexual dimorphism in body weight and canine size in primates. *Fol. Primatol.* **37,** 163–176.
Leutenegger, W. & Cheverud, J. M. (1982). Correlates of sexual dimorphism in primates: ecological and size variables. *Int. J. Primatol.* **3,** 387–402.
Leutenegger, W. & Cheverud, J. M. (1985). Sexual dimorphism in primates: the effects of size. In (W. L. Jungers, Ed.) *Size and Scaling in Primate Biology*, pp. 33–50. New York: Plenum Press.
Leutenegger, W. & Kelley, J. T. (1977). Relationship of sexual dimorphism in canine size and body size to social, behavioral, and ecological correlates in anthropoid primates. *Primates* **18,** 117–136.
Leutenegger, W. & Shell, B. (1987). Variability and sexual dimorphism in canine size in *Australopithecus* and extant hominoids. *J. hum. Evol.* **16,** 359–367.
Lieberman, D. E., Philbeam, D. R. & Wood, B. A. (1988). A probabilistic approach to the problem of sexual dimorphism in *Homo habilis*: a comparison of KNM-ER 1470 and KNM-ER 1813. *J. hum. Evol.* **17,** 503–511.
Lockwood, C. A., Richmond, B. G. Jungers, W. L. & Kimbel, W. H. (1996). Randomization procedures and sexual dimorphism in *Australopithecus afarensis. J. hum. Evol.* **31,** 537–548.
Lovejoy, C. O. (1981). The origin of man. *Science* **211,** 341–350.
Lovejoy, C. O., Kern, K. F., Simpson, S. W. & Meindl, R. S. (1989). A new method for estimation of skeletal dimorphism in fossil samples with an application to *Australopithecus afarensis*. In (G. Giacobini, Ed.) *Hominidae*, pp. 103–108. Milano, Jaka Book.
Lucas, P. W., Corlett, R. T. & Luke, D. A. (1986). Sexual dimorphism in tooth size in anthropoids. *Hum. Evol.* **1,** 23–39.
Martin, R. D., Willner, L. A. & Dettling, A. (1994). The evolution of sexual size dimorphism primates. In (R. V. Short & E. Balaban, Eds) *The Differences Between the Sexes*, pp. 159–200. Cambridge: Cambridge University Press.
McHenry, H. M. (1986). Size variation in the postcranium of *Australopithecus afarensis* and extant species of Hominoidea. *Hum. Evol.* **1,** 149–156.
McHenry, H. M. (1988). New estimates of body weight in early hominids and their significance to encephalization and megadontia in robust australopithecines. In (F. E. Grine, Ed.) *Evolutionary History of the ''Robust'' Australopithecines*, pp. 133–148. New York: Aldine de Gruyter.
McHenry, H. M. (1991). Sexual dimorphism in *Australopithecus afarensis. J. hum. Evol.* **20,** 21–32.
McHenry, H. M. (1992). Body size and proportions in early hominids. *Am. J. phys. Anthrop.* **87,** 407–431.
McHenry, H. M. (1994*a*). Behavioral ecological implications of early hominid body size. *J. hum. Evol.*
McHenry, H. M. (1994*b*). Sexual dimorphism in fossil hominids and its sociological implications. In (S. Shennan & J. Steele, Eds) *The Archaeology of Human Ancestry. Power, Sex, and Tradition.* London: Routledge and Kegan Paul. pp. 91–109.
Mitani, J. C. (1985). Territoriality and monogamy among agile gibbons (*Hylobates agilis*). *Behav. Ecol. Sociobiol.* **20,** 265–269.
Mitani, J., Gros-Lousi, J. & Richards, A. F. (1995). Sexual dimorphism, the operational sex ratio, and the intensity of male competition in polygynous primates. *Am. Nat.* **147,** 966–980.
Morbeck, M. E. & Zihlman, A. L. (1989). Body size and proportions in chimpanzees, with special reference to *Pan troglodytes schweinfurthii* from Gombe National Park, Tanzania. *Primates* **30,** 369–382.
Nishida, T. & Hiraiwa-Hasegawa, M. (1987). Chimpanzees and bonobos: cooperative relationships among males. In (B. B. Smuts, D. L. Cheney, R. M. Seyfarth, R. W. Wrangham & T. T. Struhsaker, Eds) *Primate Societies*, pp. 165–177. Chicago: University of Chicago Press.
Olsen, T. D. (1981). Basicranial morphology of the extant hominoids and Pliocene hominids: the new material from the Hadar Formation, Ethiopia, and its significance in early human evolution and taxonomy. In (C. B. Stringer, Ed.) *Aspects of Human Evolution*, pp. 99–128. London: Taylor and Francis.
Olsen, T. R. (1985). Cranial morphology and systematics of the Hadar Formation hominids and ''*Australopithecus*'' *africanus*. In (E. Delson, Ed.) *Ancestors: The Hard Evidence*, pp. 102–119. New York: Alan R. Liss, Inc.
Pilbeam, D. R. & Zwell, M. (1972). The single species hypothesis, sexual dimorphism, and variability in early hominoids. *Yearb. Phys. Anthropol.* **16,** 69–79.
Plavcan, J. M. (1990). Sexual Dimorphism in the Dentition of Extant Anthropoid Primates. Ph.D. Dissertation, University Microfilms, Ann Arbor, Michigan.
Plavcan, J. M. (1993). Canine size and shape in male anthropoid primates. *Am. J. phys. Anthrop.* **92,** 201–216.
Plavcan, J. M. (1994). Comparison of four simple methods for estimating sexual dimorphism in fossils. *Am. J. phys. Anthrop.* **94,** 465–476.

Plavcan, J. M. & van Schaik, C. P. (1992). Intrasexual competition and canine dimorphism in anthropoid primates. *Am. J. phys. Anthrop.* **87,** 461–477.
Plavcan, J. M. & van Schaik, C. P. (1994). Canine Dimorphism. *Evol. Anthropol.* **2,** 208–214.
Plavcan, J. M. & van Schaik, C. P. (1997). Intrasexual competition and body weight dimorphism in anthropoid primates. *Am. J. phys. Anthrop.* (in press).
Plavcan, J. M. & Kelley, J. (1996). Evaluating the ''dual selection'' hypothesis of canine reduction. *Am. J. phys. Anthrop.* **99,** 379–387.
Plavcan, J. M., van Schaik, C. P. & Kappeler, P. M. (1995). Competition, coalitions and canine size in primates. *J. hum. Evol.* **28,** 245–276.
Pusey, A. E. (1978). The Physical and Social Development of Wild Adolescent Chimpanzees (*Pan troglodytes schweinfurthii*). Ph.D. Dissertation, Standford University.
Richmond, B. G. & Jungers, W. L. (1995). Size variation and sexual dimorphism in *Australopithecus afarensis* and living hominoids. *J. hum. Evol.* **29,** 229–245.
Robinson, J. G. Wright, P. C. & Kinzey, W. G. (1987). Monogamous cebids and their relatives: intergroup calls and spacing. In (B. B. Smuts, D. L. Cheney, R. M. Seyfarth, R. W. Wrangham & T. T. Struhsaker, Eds) *Primates Societies*, pp. 146–154. Chicago: The University of Chicago Press.
Rodman, P. S. (1984). Foraging and social systems of orangutans and chimpanzees. In (P. S. Rodman & J. G. H. Cant, Eds) *Adaptations for Foraging in Nonhuman Primates*, pp. 134–150. New York: Columbia University Press.
Schmid, P. (1989). How different is Lucy? In (G. Giacobin, Ed.) *Hominidae: Proceedings of the 2nd International Congress of Human Paleontology*, pp. 109–114. Milan: Jaca Book.
Senut, B. (1986). Long bones of the primates upper limb: monomorphic or dimorphic? *Hum. Evol.* **1,** 7–22.
Senut, B. & Tardieu, C. (1985). Functional aspects of Plio-Pleistocene hominid limb bones: implications for taxonomy and phylogeny. In (E. Delson, Ed.) *Ancestors: The Hard Evidence*, pp. 193–201. New York: Alan R. Liss, Inc.
Shea, B. T. (1986). Ontogenetic approaches to sexual dimorphism in anthropoids. *Hum. Evol.* **1,** 97–110.
Sokal, R. R. & Rohlf, F. J. (1995). *Biometry: The Principles and Practice of Statistics in Biological Research*. 3rd edn. New York: W.H. Freeman and Co.
Stern, J. T. & Sussman, R. L. (1983). The locomotor anatomy of *Australopithecus afarensis. Am. J. phys. Anthrop.* **60,** 279–317.
Stewart, K. J. & Harcourt, A. H. (1987). Gorillas: variation in female relationships. In (B. B. Smuts, D. L. Cheney, R. M. Seyfarth, R. W. Wrangham & T. T. Struhsaker, Eds) *Primate Societies*, pp. 155–164. Chicago, University of Chicago Press.
Susman, R. L., Stern, J. T. Jr & Jungers, W. L. (1984). Arboreality and bipedality in the Hadar hominids. *Fol. Primatol.* **43,** 113–156.
Uehara, S. & Nishida, T. (1987). Body weights of wild chimpanzees (*Pan troglodytes schweinfurthii*) of the Mahale Mountains National Park, Tanzania. *Am. J. phys. Anthrop.* **72,** 315–321.
White, T. D. (1977). New fossil hominids from Laetoli, Tanzania. *Am. J. phys. Anthrop.* **46,** 197–230.
White, T. D., Johanson, D. C. & Kimbel, W. H. (1981). *Australopithecus africanus*: its phyletic position reconsidered. *S. Afr. J. Sci.* **77,** 445–470.
White, T. D., Suwa, G. & Asfaw, B. (1994). *Australopithecus ramidus*, a new species of early hominid from Aramis, Ethiopia. *Nature* **371,** 306–312.
White, T. D., Suwa, G., Hart, W. K., Walter, R. C., WoldeGabriel, G., de Heinzelin, J., Clark, J. D., Asfaw, B. & Vrba, E. (1993). New discoveries of *Australopithecus* at Maka in Ethiopia. *Nature* **366,** 261–265.
Wolpoff, M. H. (1976). Some aspects of the evolution of early hominid sexual dimorphism. *Curr. Anthropol.* **17,** 579–606.
Wood, B. A. (1993). Early *Homo*: how many species? In (W. L. Kimbel & L. B. Martin, Eds) *Species, Species Concepts, and Primate Evolution*, pp. 485–522. New York: Plenum Press.
Wrangham, R. W. (1987). The significance of African apes for reconstructing human social evolution. In (W. G. Kinzey, Ed.) *The Evolution of Human Behavior: Primate Models*, pp. 51–71. Stony Brook: State University of New York Press.
Wrangham, R. W. & Smuts, B. B. (1980). Sex differences in the behavioral ecology of chimpanzees in the Gombe National Park, Tanzania. *J. Reprod. Fert.*, **28** (Suppl.), 13–31.
Zihlman, A. L. (1985). *Australopithecus afarensis*: two sexes or two species? In (P. V. Tobias, Ed.) *Hominid Evolution: Past, Present, and Future*, pp. 213–220. New York: Alan R. Liss.